

Assets and Links

Figma Prototype : <https://www.omnicrm.expomore.in>

Nexacalling : <https://nexacalling.com/login>

Define business logic

System and process flows - in mermaid format for draw.io

Validations, constraints, and edge case document

Storage

Backup

Recovery

MODULE 1 — Tenant & License Management

SRS v1.0

AI Calling SaaS + CRM Platform

Status: Approved Discovery → Structured Specification

MODULE 1 — Tenant & License Management

Tenant → Customer

1. Purpose

Enable a **license-driven, multi-tenant SaaS** where each customer operates independently, with strict usage enforcement tied to AI calling consumption.

This module is the **commercial backbone** of the platform.

2. Architectural Assumptions (Frozen)

- **! Dedicated Database per Tenant**
 - Each customer has an isolated database instance.
 - No shared DB, no `tenant_id` logic.
 - Improves data security, compliance, and enterprise sales readiness.
- **! Platform-Owned Telephony**
 - Platform owns numbers, providers, and routing.
 - Customers cannot bring their own telephony credentials.
 - Billing is fully controlled via minute consumption.
- **! Usage-Based Licensing**
 - feature tiers and Pricing - customisable
 - No white-label.- yes
 - Pricing = **per-minute AI call usage**.
 - License duration configurable (monthly / yearly / custom).

These assumptions are now **locked** unless you explicitly request a change.

3. Functional Requirements

3.1ne]

- Create tenant with **Tenant Lifecycle Management** [Doh:
 - Company name
 - Primary contact details - address, city, contact person . mno. telephone , pan number
 - KYC - Auto fetch required with country, Pan Card / pan certificate . company incorporation certificate , Company related document.
 - User can put name of document as well as upload the document.
 - Timezone
 - Locale
 - Currency
 - Industry (optional, for analytics)
 - Tenant status:
 - Active
 - Suspended
 - Expired
 - Locked
 - Soft delete with retention period (configurable)
-

3.2 License Configuration[Done]

- License attributes:
 - Number of companies
 - License type: trial/Monthly/quarterly/half yearly / Yearly / Custom
 - Number of users based
 - Concurrent users
 - Named users
 - Credit based on features
 - License Number Format : auto generate (logic required - sujat and akash)
 - Feature and module configured
 - Ai calling
 - Whatsapp
 - Partner Name
 - Referral name
 - Team → Assigned staff
 -
- License can be:
 - Assigned
 - Renewed
 - Suspended
 - Terminated
- License changes **take effect immediately** or **on next billing cycle** (admin-controlled)

License request form system : TO BE Discussed with Akash sir

Finance approval [Not Done]

- Payment status (amount received , due , partial and full, receipt number) and billing status - bill date, bill amount, bill number .
- Data migration to the accounting software (post to Accounting as sales order) accountant will convert into bill.
- Order → agreement → payment → tax invoice → final license (before that temp license)
- Temp lis - trial → temp- later converted to permanent license and demo

Management approval [Not done]

- License agreement generation → format and to be sent on email, whatsapp.
- Dual Authentication basis approval for license (Maker and checker)

User onboarding [Done]

Trial license → payment through payment gateway → Auto approval

License payment → Online or offline [only design]

1. Issue Temp invoice for 15 days
2. Licenses generate
3. Tax invoice
4. Customer pay
5. Customer submit proof
6. Then go for approval for admin

License, Billing and Payment status[Not done]

Tenant based configuration [not done]

1. Database (MySQL) →
2. Domain name

License plan – upgrade and degrade [Done , but not active]

Upgrade : new license created

Same for degrade

Payment Settlement :

1. Pendrive : unique ID , license number ,

Subscription plan needs to design

Software license and payment cycle → degradation is not possible upto license time.

3.3 Usage Tracking & Metering[partially done/changes requested]

Tracked at **real-time granularity**:

- Credits (primary billable metric)
- Number of AI agents created
- Concurrent calls (for operational limits)
- CRM record count (for analytics only, not billing)
- Number of companies
- Number of users (pricing will be added accordingly and configurable for on and off)
- Tier pricing - configurable by super admin

The screenshot displays the Nexacalling Admin Dashboard. On the left is a sidebar with navigation links: Forms, Appointments, Templates, TOOLS (Webhooks, Website Widget), BILLING (Upgrade Plan, Billing & Credits, Transaction History), and ADMINISTRATION (Admin Dashboard). A 'Credits' widget shows 9,980. The main area is titled 'Plan Management' with the subtitle 'Configure subscription plans and their limits'. A 'Create New Plan' form is open, with the instruction 'Add a new subscription plan for users'. The form fields include: Plan Name (Internal) with value 'e.g., premium', Display Name with value 'e.g., Premium Plan', Description with value 'Plan description', Monthly Price (USD) with value 0, Yearly Price (USD) with value 0, Max Agents with value 1, Max Campaigns with value 1, Max Contacts with value 5, Max Webhooks with value 3, Max Knowledge Bases with value 5, Max Flows with value 3, Max Phone Numbers with value 1, Max Widgets with value 1, and Included Credits with value 100. The bottom of the screen shows a Windows taskbar with the date 22-01-2026 and time 14:48.

- Module wise usage

Usage data must be:

- Immutable
- Auditable

- Time-stamped
-

Pay-as-you-Go method[Not done]

Components : whatsapp, ai calling, sms , AI features , email , license renewal, AI Chat Bot,

Add-on module Method

1. Accounting
2. HRMS
- 3.

3.4 License Enforcement Rules[not done]

- **Hard Enforcement (Confirmed) : when features will be planned**
 - When Credit reaches zero:
 - Notification on custom credit alert - whatsapp and email
 - AI calls are blocked
 - Campaigns auto-paused
 - Inbound calls receive system message
 - Date expiry - no access
- **Soft Alerts**
 - Configurable thresholds (e.g., 80%, 90%)
 - Email + in-app alerts to tenant admin

No grace usage. No silent overages.

- Can be added through admin on approval
-

3.5 Tenant Configuration Scope

Each tenant can configure:[done]

- Business hours
 - Default call timezone
 - Data retention policies (within compliance limits)
 - Notification preferences
 - Low level configuration system
 - - take format from Akash ji
-

4. Admin Controls (Module Scope)

Super Admin can:[done]

- Create / edit / suspend tenants
 - OTP based login
 - Assign or override licenses
 - View usage across all tenants
 - Force-stop campaigns
 - Lock accounts for compliance violations on 2FA
-

5. In Scope

- White-labeling
- Feature-tier licensing
- Per-seat pricing

These are intentionally excluded to protect **business simplicity and margin clarity**.

Configuration modules to be defined

NON-FUNCTIONAL REQUIREMENTS (Baseline v1.0)

These apply **across all modules**.

1. Performance

- API response time:
 - $\leq 300\text{ms}$ (95th percentile)
 - AI call initiation latency:
 - ≤ 2 seconds
 - System must support:
 - Minimum 1,000 concurrent calls (scalable)
-

2. Scalability

- Horizontal scaling for:

- AI calling workers
 - Campaign processors
 - Each tenant DB must scale independently.
 - No single-tenant performance degradation.
-

3. Availability

- Target uptime: **99.9%**
 - Planned maintenance windows supported
 - Automatic failover for critical services
-

4. Security

- Role-based access enforcement
 - Encrypted data at rest & in transit
 - Call recordings encrypted
 - No cross-tenant data access under any scenario
-

5. Compliance

- GDPR-ready:
 - Data export
 - Right to delete
 - Explicit call recording consent logging
 - Configurable data retention rules per tenant
-

6. Audit & Observability

- System logs:
 - License changes
 - Usage spikes
 - Admin overrides
 - Tenant-level audit trail
 - Internal monitoring dashboards
-

7. Maintainability

- Modular services
 - Config-driven behavior (not code-driven)
 - Backward-compatible schema changes
-

Module 1 - V1.1

MODULE 1 — Tenant & License Management

Document Type: Core Platform SRS

System: Omni-Channel CRM (AI –Led)

Audience: Product, Backend, DevOps, Billing, Compliance, QA

Status: Frozen Architecture Assumptions Applied

1. Module Overview & Business Objective

1.1 Purpose

This module enables a **license-driven, multi-tenant SaaS architecture** where each customer (tenant) operates in complete isolation, and **all platform access is strictly governed by license state and AI usage consumption**.

This module is the **commercial backbone** of the platform:

- Every call, feature, and workflow ultimately resolves to license validity.
 - Revenue leakage prevention is a non-negotiable goal.
 - Compliance, auditability, and enterprise readiness are first-class citizens.
-

2. Architectural Assumptions (Frozen & Non-Negotiable)

2.1 Tenant Isolation Model

- **Dedicated database per tenant**
- No shared schema
- No `tenant_id`-based row isolation
- **One database = One Entity/ Company**
- **Max company has multiple database - because data is slow (max 10-15 GB) Logs**

Rationale

- Eliminates cross-tenant data bleed risk
- Simplifies compliance (SOC2, ISO, HIPAA, GDPR -adjacent use cases)
- Nepali date , Audit trail, no direct deletion
- Strengthens enterprise sales posture

2.2 Platform-Owned Telephony

- Platform owns:
 - Phone numbers
 - Telephony providers
 - Routing logic
- Tenants **cannot** bring external telephony credentials
- All billing flows through platform-controlled metering

Rationale

- Prevents revenue bypass
 - Guarantees consistent QoS
 - Simplifies enforcement and analytics
-

2.3 Licensing & Pricing Model

- Usage-based licensing
- Primary billable metric: **Credits**
- License duration:
 - Monthly
 - Yearly
 - Custom (date-based)
- Feature tiers are configurable
- **White-labeling: explicitly excluded**

These assumptions are locked unless explicitly revised via change request.

3. Functional Requirements

3.1 Tenant Lifecycle Management

3.1.1 Tenant Creation

System shall allow Super Admin to create a tenant with the following attributes:

Mandatory Fields

- Company Legal Name
- Registered Address
- City
- Country
- Primary Contact Person Name
- Primary Contact Mobile Number
- Primary Contact Email
- Timezone
- Locale
- Currency

Compliance & KYC Fields

- PAN Number
- PAN Certificate (upload)
- Certificate of Incorporation (upload)

Optional Metadata

- Industry (used only for analytics and reporting)
-

3.1.2 Tenant Status Lifecycle

Each tenant shall exist in exactly one of the following states:

Status	Description
Active	Full access as per license
Suspended	Temporarily disabled by admin
Expired	License validity ended
Locked	Compliance or security lock
Soft Deleted	Hidden but retained for recovery

Rules

- Soft-deleted tenants are retained for a configurable retention period
- Hard deletion is a backend-only administrative action
- Status changes must be logged with:

- Timestamp
 - Actor (admin/system)
 - Reason code
-

3.2 License Configuration & Management

3.2.1 License Attributes

Each tenant shall have at least one active license record containing:

- License Type: Monthly / Yearly / Custom
- License Validity Start Date
- License Validity End Date
- Total Allowed Credits
- Feature Modules Enabled:
 - AI Calling
 - WhatsApp
 - (Future modules configurable)
- Concurrent Call Limit
- Agent Creation Limit
- User Limit (configurable, pricing TBD)

Datta: Not fully functioning with Link showing details of users.. → Its done please check

3.2.2 License Number Generation

- License number must be **auto-generated**
 - Format must be deterministic and unique
 - Logic must support prefix customization (internal naming logic to be finalized separately)
-

3.2.3 License State Transitions

License may transition between:

- Assigned
- Renewed
- Suspended
- Terminated
- Expired

Rules

- Admin may choose:
 - Immediate effect
 - Next billing cycle effect
 - All transitions must be auditable
-

3.3 Usage Tracking & Metering

3.3.1 Metered Metrics

System shall track the following metrics at **real-time granularity**:

Billable

- Credits (primary revenue driver)

Operational Limits

- Number of AI agents created
- Concurrent calls
- Active campaigns

Analytics-Only (Non-Billable)

- CRM record count
- Number of companies
- Number of users

Datta: Not fully functional/Usage analysis to be modifies

3.3.2 Usage Data Integrity[after implementation]

All usage data must be:

- Immutable
 - Time-stamped
 - Non-editable post-recording
 - Queryable for audit and billing reconciliation
-

3.3.3 Tier Pricing Configuration

- Tier definitions configurable by Super Admin
- Tier changes do not retroactively affect past usage
- Future usage follows new tier rules

Datta: subscription plan to be editable..

3.4 License Enforcement Rules

3.4.1 Hard Enforcement (Confirmed Behavior)

When **AI minute balance reaches zero** OR **license expires**:

- Outbound AI calls are blocked
- Active campaigns are auto-paused
- Inbound calls:
 - Routed to system message (configurable)
- Tenant UI access restricted (date expiry case)

No silent overages permitted.

Nothing functional Archive data to be maintained for a month

Notifications not active.

Dashboard to be modified on different modules/functions.will share link..

KYC implementation

3.4.2 Notifications & Alerts

Threshold-Based Alerts

- Configurable thresholds (e.g., 80%, 90%)
- Channels:
 - Email
 - WhatsApp
 - In-app notification

Zero Balance Alerts

- Immediate notification
 - Custom credit alert messaging supported
-

3.4.3 Grace Usage Policy

- No default grace usage
 - Temporary extension allowed **only** via Super Admin approval
 - All exceptions must be logged with reason and expiry
-

3.5 Tenant-Level Configuration Scope

Each tenant admin shall be able to configure:

- Business hours
- Default call timezone
- Notification preferences
- Data retention settings (within compliance constraints)

Changes apply immediately unless specified otherwise.

4. Super Admin Controls

Super Admin shall have the ability to:

- Create, edit, suspend tenants
- Assign, override, or revoke licenses
- View aggregated usage across all tenants
- Force-stop campaigns globally
- Lock tenant accounts for compliance violations
- Login secured via OTP + 2FA

All admin actions must be:

- Logged
 - Time-stamped
 - Actor-attributed
-

5. Explicitly In Scope

The following are **intentionally excluded** from this module:

- White-labeling
- Feature-tier upsell via tenant UI
- Per-seat pricing logic

These exclusions are strategic to:

- Reduce system complexity
 - Protect revenue clarity
 - Avoid downstream support chaos
-

6. Configuration Modules (To Be Defined Separately)

The following configuration modules will be defined in subsequent SRS documents:

- Pricing Engine Configuration
 - Notification Template Management
 - Credit Top-Up & Wallet System
 - Compliance & Retention Policy Engine
-

Notes by Shivam

1. Custom Dashboard
2. Content management

Notes for Datta Sir

1. What would be parameters of search ?
2. Leave analytics for now - it would be more customisable later module
3. Verify user logs
4. Review tenant management
- 5.

1. Usage of each tenant [No of hours spent/remaining] → PUT User Based filters
2. Recently onboarded tenants/details. → On the dashboard
3. Recent activity/at user level to have much details/call wise...when clicked on user
4. License management/when clicked on license number should show more details in separate window/pop-up → clickable like tenant ID
5. Auto license deactivation on plan expiry and notification to tenant. → need complete notification list from data sir
6. Usage analysis/ when clicked on new tenants should have more details/tenant wise usage also to be implemented .. → will be taken in another module
7. Manage tenant menu option / create/delete/modify...etc.. → should be done
8. New tenant search option → search logic will be required from HITECH team
9. Payment gateway no implementation in thecredit card/debit card can be implemented. → Need to decide PG and Payment related discussion would be in another module
10. Content management....customize dashboard with pre defined layouts/designs → Dynamic Dashboard and configurable KPIs required
11. Analytics/Alters to be discussed later[Auto payment Notification , Tech support, Payment notification from Bank]
→ To be done later
12. Settings to be done later
→ settings to be done at the last
13. Usage analytics to be based on reports/filters/plugin & play[drag & drop] → it would come in analytics module
14. More interactive dashboard/Usage analysis/Individual Tenant
→ TO be DONE
15. Admin dashboard to have sales/revenue/business analysis/ Customer ratings/Generated leads[weekly/monthly/filter based]/top products sale / volume→ this would be dynamic as well

Business Playground

Business Playground

Lead Generation

ChatGPT format style : chat + predefined options + question and answer based generation

Option based

Knowledge Base : format : images, document, pdf, excel sheet

Asset management : product logo, images,

Storage capacity : per license based

About company :

1. Research and survey
 - a. Ex : electric product (kettle) : Competitor analysis , product cost, manufacturer, SWOT, Location, price, local and import, category wise- online and offline , product variants and specifications.
 - b. Chat format : AI and Internet source (pre-defined prompt, options 4-5 layers)
 - c. Visualisation process → later on
 - d. Curated/ Partnership or self data → later on
 - e. One credit per query - 1 credit/query
- Response management : history , archive, delete, add user, tag campaign
2. Strategy planning (3-4 layer options)
 - a. Campaign planning
 - b. Operational plan
 - c. Production plan
 - d. Reach strategy
3. **Data scrapping**
 - a. Use of **Scrapping Tool (assigned to akash)** to extract data as per user requirement
 - b. Export format : excel and transfer to lead management module - raw
4. Content creation
 - a. Email content
 - b. Whatsapp content
 - c. SEO
 - d. Website
 - e. Blog
 - f. Image
 - g. Video

- h. Reels
- i. AI calling script
- j. Bulk calling script
- k. Follow up messages/emails

5. Social media marketing - reference sinatra.ai - agentic format

- a. Auto posting
- b. Data fetch from ad campaign
- c. Ad management : optional

6. Direct marketing → email, whatsapp, RCS, SMS

- a.

7. AI Calling marketing

8. Bulk calling

9. Task assignment - manual working

Business PG

Module: Business Playground & Lead Generation Suite

Platform: CRM System

Prepared By: TechAvtar

1. Module Overview

The **Business Playground & Lead Generation Suite** is an AI-powered research, strategy, and outreach engine embedded within the CRM.

It enables businesses to:

- Generate qualified leads
- Conduct company/product research
- Build GTM strategies
- Plan campaigns & operations
- Create marketing content
- Scrape & enrich external data
- Execute outreach via multi-channel marketing

The module will operate in a **chat-driven + option-based workflow**, allowing both guided and AI-generated interactions.

2. Core Functional Architecture

2.1 AI Chat & Guided Interaction Engine

Format Style:

- Conversational chat interface
- Predefined option layers (5 levels deep)
- Question → Answer → Insight generation
- AI + Internet source blended responses

Capabilities:

- Prompt templates (pre-defined business queries)
- Multi-step guided research flows
- Industry-specific question trees
- Lead discovery queries
- Strategy recommendations

Credit Logic:

- 1 Credit = 1 Query execution
 - Credit deduction per AI + research response
-

3. Knowledge Base & Asset Management

3.1 Knowledge Repository

Supported formats:

- Images
- PDF documents
- Excel sheets
- Word files
- Research datasets

Functions:

- Upload / Import knowledge assets
 - Tagging & categorization
 - Searchable indexing
 - AI referencing during responses
-

3.2 Asset Management System

Asset types:

- Product logos
- Brand creatives
- Marketing images
- Campaign banners
- Product packaging

Features:

- Folder structuring
- Metadata tagging
- Campaign mapping
- Reuse in content creation

Storage Model:

- License-based storage allocation
 - Tiered storage expansion
 - Admin quota monitoring
-

4. Company & Product Research Engine

4.1 Research & Survey Module

Example Use Case: *Electric Kettle Market Research*

Research coverage:

- Competitor analysis
 - Product cost benchmarking
 - Manufacturer identification
 - SWOT analysis
 - Pricing comparison
 - Import vs local sourcing
 - Category segmentation
-

4.2 Data Dimensions

Market Intelligence Layers:

1. Online marketplaces
2. Offline retail
3. Regional distributors
4. Import/export vendors

Product Analysis:

- Variants & SKUs
 - Technical specifications
 - Certifications
 - Packaging formats
 - Warranty & service models
-

4.3 Data Source Types

- AI-generated insights
 - Internet research aggregation
 - Curated partner datasets (*Phase 2*)
 - Proprietary datasets (*Phase 3*)
-

5. Strategy Planning Engine

A multi-layer guided planning system for business growth.

5.1 Planning Layers (3–4 Levels Deep)

1 Campaign Planning

- Objective definition
- Target audience
- Budget planning
- Platform selection

2 Operational Planning

- Team allocation
- Timeline structuring
- Execution workflows

3 Production Planning

- Vendor sourcing
- Manufacturing timelines
- Cost planning

4 Reach Strategy

- Channel mapping
 - Distribution planning
 - Geographic targeting
-

6. Lead Generation & Data Scraping

6.1 Scraping Engine Integration

- External scraping tools integration
- Assigned scraping operations (internal team mapping e.g., Akash)
- Requirement-based extraction

Extractable Data:

- Company lists
 - Decision makers
 - Emails & phones
 - Product distributors
 - Retailer databases
-

6.2 Data Processing Flow

Scraped data → Export → Lead ingestion

Export Formats:

- Excel
- CSV
- API push

CRM Integration:

- Direct transfer to Lead Management Module
 - Raw lead tagging
 - Source attribution
-

7. Response & Data Management

7.1 Query Response Management

Features:

- Query history tracking
 - Archive responses
 - Delete records
 - Re-run queries
 - Campaign tagging
-

7.2 User & Access Controls

- Multi-user access
 - Role-based permissions
 - Credit usage monitoring
-

8. AI Content Creation Suite

A built-in generative content engine for outreach and branding.

8.1 Content Types

Text Content

- Email campaigns
- WhatsApp messages
- Follow-up sequences
- Sales scripts

Marketing Content

- SEO articles
- Website copy
- Blogs

Creative Content

- Images
- Ad creatives
- Social posts

Video Content

- Reels
- Promo videos
- Product explainers

Voice & Calling

- AI calling scripts
 - Bulk call scripts
 - IVR flows
-

9. Social Media Marketing Engine

Reference architecture inspired by **agentic automation platforms** (e.g., Sinatra-style AI agents).

9.1 Features

- Automated post generation
 - Multi-platform publishing
 - Caption AI generation
 - Hashtag optimization
-

9.2 Campaign Integrations

- Ad campaign data fetch
 - Performance syncing
 - Lead attribution mapping
-

9.3 Auto-Posting Engine

- Scheduled publishing
 - Bulk posting
 - Campaign calendar
-

9.4 Ad Management (Optional Scope)

- Ad creation
 - Budget allocation
 - Performance tracking
 - ROI dashboards
-

10. Visualization Layer (Future Scope)

Planned analytics dashboards:

- Market research charts
 - Competitor positioning graphs
 - Pricing heatmaps
 - Lead funnel visualization
-

11. Credit & Licensing Model

Feature	Logic
Query execution	1 Credit per query
AI research depth	Credit-weighted
Scraping jobs	Premium credits
Content generation	Credit-based

12. Module Deliverables

- AI Chat Research Interface
 - Knowledge Base Repository
 - Asset Management System
 - Product Research Engine
 - Strategy Planner
 - Lead Scraping Integration
 - Lead Export & CRM Sync
 - Content Creation Studio
 - Social Media Automation
 - Response History Manager
-

MODULE 3 — Lead Management (CRM Core)

MODULE 3 — Lead Management (CRM Core)

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module is the **system of record** for all prospects and customers.

Every AI call, human action, campaign, and deal must anchor back to this data model.

Bad CRM design kills AI effectiveness. This module is designed to **serve AI first, humans second**—without sacrificing usability.

2. Scope

Applies to:

- Lead capture
- Lead enrichment
- Assignment & routing
- Lifecycle management
- AI-triggered actions
- Human follow-ups

Does **not** include:

- Deal revenue tracking (covered in Module 6)
 - Billing data
 - External CRM sync (optional future)
-

3. Lead Entity Definition

3.1 Core Lead Attributes

Mandatory:

- Lead ID (system generated)
- Name
- Primary phone number
- Source
- Status / Stage
- Owner (user or unassigned)
- Created timestamp

Optional / Enrichable:

- Email
 - Alternate phone
 - Company
 - Location
 - Language preference
 - Tags
 - Custom fields (tenant-defined) → validation, text, number, date and time, attachment, image, single choice and multiple choice .
-

4. Lead Sources (Confirmed)

- Web forms
- CSV import
- API ingestion
- WhatsApp
- SMS
- Manual entry
- AI-call generated leads
- Meta
-

Each lead must store:

- Source type
 - Source metadata
 - First-touch timestamp
-

5. Lead Lifecycle & Pipelines

5.1 Pipelines

- Multiple pipelines per tenant (confirmed)
 - Pipelines are:
 - Fully configurable
 - Independent per business use case
 - Example:
 - Sales Pipeline
 - Support Intake
 - Collections
 - Surveys
-

5.2 Lead Stages

- Tenant-defined stages
 - Each stage supports:
 - Entry conditions
 - Exit conditions
 - Allowed actions
 - Stage changes are:
 - Logged
 - Timestamped
 - Attributable (AI or user)
-

6. Lead Assignment Rules

6.1 Assignment Modes

- Manual assignment
- Round-robin
- Rule-based (conditions)
- AI-assisted routing

6.2 Assignment Triggers

- On lead creation
 - On stage change
 - On AI call outcome
 - On inactivity timeout
-

7. Lead Scoring (Confirmed: Hybrid)

7.1 Rule-Based Scoring

- Configurable rules:
 - Source weight
 - Call outcome
 - Engagement level
 - Score updates in real time
-

7.2 AI-Based Scoring

- AI evaluates:
 - Call transcript sentiment
 - Intent strength
 - Objection patterns
 - Score confidence stored separately
-

8. AI ↔ Lead Interaction Rules

8.1 AI Permissions on Leads

AI is allowed to:

- Update lead status
- Add notes
- Schedule follow-ups
- Trigger campaigns
- Create tasks

AI is **not allowed** to:

- Delete leads
 - Merge leads
 - Modify billing-related fields
-

8.2 Follow-up Logic

- AI can:
 - Schedule callbacks
 - Trigger WhatsApp/SMS/email
 - Escalate to human
 - Follow-ups are stored as first-class objects
-

9. Duplicate Detection

9.1 Detection Rules

- Primary phone number (strict)
- Secondary phone / email (soft)
- Fuzzy name matching (optional)

9.2 Duplicate Handling

- Auto-merge (configurable)
 - Flag for review
 - Link as related leads
-

10. Activity Timeline

Each lead has a **single chronological timeline** containing:

- AI calls
- Human calls
- Notes
- Status changes
- Campaign touches
- Messages sent
- Manual log entry

Timeline is:

- Read-only
 - Immutable
 - Filterable
-

11. Custom Fields & Extensibility

- Tenant admins can define:
 - Field type → explained above
 - Validation rules
 - Required/optional
 - Fields are:
 - Indexed
 - Searchable
 - Available to AI context
-

12. Search & Filtering

- Global search
- Advanced filters:
 - Stage
 - Score range
 - Last activity
 - Assigned user
- Saved views per user

- Tags
 - Link → link to another existing software
 - Project/Class Number
 - Authorisation and approval
-

13. Non-Functional Requirements (Module-Specific)

- Lead read/write latency $\leq 200\text{ms}$
 - Search response $\leq 500\text{ms}$
 - Must support:
 - ≥ 10 million leads per tenant
 - No hard dependency on AI availability
-

14. Out of Scope (Explicit)

- Revenue calculations
 - Third-party CRM sync
 - Marketing email campaigns
-

Module 3 - V1.1

MODULE 3 — Lead Management (CRM Core)

Version: 1.0

Status: Draft → Pending Confirmation

System: Omni-Channel CRM Platform

Audience: Product, Backend, AI, QA, Analytics

1. Module Overview & Design Intent

1.1 Purpose

This module is the **authoritative system of record** for all prospects, inquiries, and customer touchpoints.

Every downstream system must anchor here:

- AI calls
- Human follow-ups
- Campaign execution
- Scoring, routing, analytics
- Opportunity creation (future module)

Design principle:

Bad CRM data kills AI effectiveness.

This module is **AI-first by design, human-usable by discipline.**

2. Module Scope

2.1 In Scope

- Lead capture & ingestion
- Lead enrichment (human + AI)
- Assignment & routing
- Lifecycle & pipeline management
- AI-triggered actions
- Human follow-ups & activities/task
- Central activity timeline

2.2 Explicitly Out of Scope

- Deal / revenue tracking (Module 6)
 - Billing & invoices/agreement
 - External CRM synchronization (future enhancement)
-

3. Lead Entity Definition (Core Data Model)

3.1 Lead Identity

Each lead must be uniquely identifiable and immutable.

System-Generated

- Lead ID (globally unique per tenant)
 - Created timestamp
 - Created by (**User ID / AI Agent ID / System**)
-
1. **User** : Manual single entry
 2. **User** : Manual bulk upload or entry through sheet and CSV
 3. **AI** : Agentic lead generation - single or bulk
 4. **System** : **System connected to social ad management** (ex: meta, indiamart, alibaba, google, whatsapp, forms, ecommerce - shopify, woocommerce)
 5. **System** : Manual connect with **Webhook** and automation
- Shivam to do : data mapping , Akash ji explore import in hubspot

Total lead captured : 1000

Campaign : location wise, user, product wise , Multiple

Cam 1 : Product wise - jewellery software : 300 leads

Cam 2: product wise : trading software : 500 leads

Cam 3 : product wise : production software : 200 leads

3.2 Default Mandatory Attributes

- Full Name
 - Primary Phone Number
 - Lead Source
 - Pipeline
 - Current Stage
 - Owner (User or Unassigned)
 - Status (Open / Closed / Archived)
-

3.3 Optional & Enrichable Attributes

- Email address
 - Alternate phone number(s)
 - Company name
 - Location (city, state, country)
 - Preferred language
 - Tags (multi-value)
 - Notes (structured + free text)
 - Gst
 -
-

3.4 Custom Fields (Tenant-Defined)

Tenant Admins may define custom lead fields with the following types:

- Text
- Number
- Date
- Date & Time
- Attachment
- Image
- Single Choice
- Multiple Choice
- Validation-based fields

Rules

- Custom fields must support:
 - Validation rules
 - Required/optional flags

- Fields must be:
 - Indexed
 - Searchable
 - Available to AI context ingestion
-

4. Lead Sources (Confirmed & Expandable)

Supported lead sources include:

- Web forms
- CSV import
- API ingestion
- WhatsApp
- SMS
- Manual entry
- AI-call generated leads
- Meta platforms (ads / forms)
- LinkedIn

Each lead must persist:

- Source type
 - Source metadata (campaign, channel, ref ID)
 - First-touch timestamp
-

5. Lead Lifecycle & Pipelines

5.1 Pipelines

- Multiple pipelines per tenant (confirmed)
- Pipelines are fully configurable
- Each pipeline represents a **business workflow**

Examples (aligned with client CRM reference):

- Sales / Inquiry Pipeline
- Support Intake
- Collections
- Surveys / Feedback

5.2 Stages

Each pipeline consists of tenant-defined stages.

Each stage supports:

- Entry conditions
- Exit conditions
- Allowed actions
- Automation hooks (AI / workflow)

Stage transitions must be:

- Logged
 - Call logs with all metadata - time, recordings (Mobile APP) : reference : <https://telecrm.in/>
 - Reachout methods
 - Whatsapp chat interface
 - Email interface
 - Wechat integration ?
 - SMS logs
- Timestamped
- Attributed (AI or user)
 - AI Agent
 - Human Agent

Workflow Automation

Tools : n8n, clawbot

UX and User friendly automation and condition based execution.

Workflow stage level

Workflow stage validation level

6. Lead Assignment & Routing

6.1 Assignment Modes

- Manual assignment
- Round-robin

- Rule-based (conditions)
 - AI-assisted routing
-

6.2 Assignment Triggers

Assignment may occur:

- On lead creation
- On stage change
- On AI call outcome
- On inactivity timeout (SLA-based)

All assignments must be logged with:

- Previous owner
 - New owner
 - Trigger source → n8n
 - Timestamp
-

7. Lead Scoring (Hybrid Model — Confirmed)

7.1 Rule-Based Scoring

Configurable scoring rules include:

- Lead source weight → every stage weight manual and AI
- Call outcome
- Engagement frequency
- Response time

Scores update in near real time.

7.2 AI-Based Scoring

AI evaluates:

- Call transcript sentiment
- Intent strength
- Objection patterns

- Conversation depth

Rules

- AI score confidence stored separately
 - AI score does not overwrite rule-based score
 - Final score is a composite (configurable weightage)
-

8. AI ↔ Lead Interaction Rules

8.1 AI Permissions on Leads

AI agents are allowed to:

- Update lead status/stage
- Add structured notes
- Schedule follow-ups
- Trigger campaigns
- Create tasks

AI agents are **not allowed** to:

- Delete leads
- Merge leads
- Modify billing-related or compliance fields

Enforcement is mandatory at API level.

8.2 Follow-Up Logic

AI can:

- Schedule callbacks
- Trigger WhatsApp / SMS / Email
- Escalate to human users

Follow-ups are stored as **first-class entities**, not notes.

9. Duplicate Detection & Handling

9.1 Detection Rules

- Primary phone number → strict match
 - Secondary phone / email → soft match
 - Fuzzy name matching → optional
-

9.2 Duplicate Handling Options

Configurable per tenant:

- Auto-merge
- Flag for review
- Link as related leads

All merge actions must be logged and reversible (where possible).

- Optional knowledge based inclusion at the time campaign
-

10. Activity Timeline (Single Source of Truth)

Each lead must have **one unified chronological timeline** containing:

- AI calls
- Human calls
- Notes
- Status & stage changes
- Campaign touches
- Messages sent
- Manual log entries

Timeline Properties

- Read-only
- Immutable
- Filterable
- System-generated ordering only

This replaces scattered “activity tables” from legacy CRM designs.

11. Search, Filters & Views

11.1 Search

- Global lead search
 - Indexed across standard + custom fields
-

11.2 Filters

Advanced filtering by:

- Pipeline
 - Stage
 - Score range
 - Last activity date
 - Assigned user
 - Tags
-

11.3 Saved Views

- Users may save personal views
 - Tenant Admins may define shared views
-

12. Integrations & References

12.1 Linking & External References

Leads may store:

- External system links
- Project / Class numbers
- Authorization & approval references

These fields are non-billable metadata and AI-readable.

13. Security & Audit Constraints

- No cross-tenant visibility
 - All lead mutations must be attributable
 - Role-based permissions from Module 2 apply strictly
 - All critical actions logged immutably
-

MODULE 4 — AI Calling Engine

MODULE 4 — AI Calling Engine (Core Differentiator)

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module is the **engine of value creation**.
Everything else supports it.

The AI Calling Engine must:

- Sound human
- Act intelligently
- Stay controllable
- Never lose cost discipline

This is not a chatbot with a phone number. This is a **stateful, goal-driven voice system tightly bound to CRM data**.

2. Scope

Applies to:

- Outbound AI calls
- Inbound AI calls
- Conversation handling
- CRM updates via voice
- Human handoff
- Failure recovery

Does **not** include:

- Telephony vendor management UI
- Billing logic (Module 9)
- Post-call analytics dashboards (Module 7)

3. Call Types (Confirmed)

3.1 Supported Call Directions

- Outbound calls
- Inbound calls

Both must support:

- AI-first handling
 - Human fallback
 - CRM context injection
-

4. AI Agent Model

4.1 AI Agent Types (Confirmed)

- Sales Agent
- Support Agent
- Collections Agent
- Survey Agent

Each AI agent type is:

- Configurable
 - Reusable
 - Deployable across campaigns
-

4.2 AI Agent Identity

Each AI agent has:

- Agent ID
- Agent type
- Language(s)
- Voice profile
- Personality tone
- Default objectives

AI agents are **tenant-scoped**, not user-scoped.

5. Conversation Intelligence

5.1 Core Capabilities (Confirmed)

- Natural turn-taking
- Interrupt handling
- Backchanneling (acknowledgments)
- Context carryover within call
- Intent detection
- Objection handling

No scripted IVR behavior unless explicitly configured.

5.2 Context Sources

AI must consume context from:

- Lead data
- Previous call summaries
- Campaign instructions
- Tenant rules
- Time & timezone

Context injection must be **deterministic and auditable**.

6. Conversation Design (Confirmed: Hybrid)

6.1 Prompt-Based Design

- System prompt
- Goal prompt
- Guardrails
- Compliance instructions

Used for:

- Fast setup
 - Simple campaigns
-

6.2 Flow-Based Design

- Node-based conversation flows
- Conditional branching
- Data capture nodes
- Action nodes

Used for:

- Regulated flows
- High-stakes calls
- Enterprise customers

Both approaches can coexist per tenant.

7. Call Lifecycle State Machine

Every call must follow a **strict state model**:

1. Initialized
2. Dialing / Ringing
3. Connected
4. Active Conversation
5. Action Execution
6. Wrap-up
7. Completed / Failed

State transitions are logged and immutable.

8. Actions Executable by AI

AI is allowed to:

- Update lead fields
- Change lead stage
- Create notes

- Schedule callbacks
- Trigger WhatsApp/SMS follow-ups
- Transfer to human agent
- End call intentionally

AI is **not allowed** to:

- Modify billing data
 - Delete records
 - Override compliance rules
-

9. Fallback & Failure Handling (Confirmed)

9.1 Failure Scenarios

- ASR failure
 - LLM failure
 - Tool execution failure
 - Silence / no response
 - User frustration detected
-

9.2 Fallback Actions (Confirmed)

- Retry (configurable attempts)
- Transfer to human
- Drop call & tag lead
- Auto-schedule follow-up

Fallback behavior is:

- Campaign-specific
 - Cost-aware
 - Logged
-

10. Human Handoff

10.1 Transfer Rules

- Warm transfer (AI summarizes context)
 - Cold transfer (optional)
 - Agent availability checks
-

10.2 Post-Transfer Behavior

- AI stops billing immediately
 - Call ownership switches to human
 - Conversation summary auto-generated
-

11. Multi-Language Support

- Language auto-detection (optional)
- Language forced per campaign
- Voice + prompt consistency required

Language changes mid-call must be explicit.

12. Call Recording & Transcription

- All calls recorded (configurable)
 - Transcripts generated post-call
 - Stored per lead timeline
 - Consent captured and logged
-

13. Cost Control & Safeguards

Critical requirement.

- Max call duration per campaign
- Silence timeout
- Repetitive loop detection
- Forced termination rules

No call should run endlessly.

14. Non-Functional Requirements (Module-Specific)

- Call setup latency ≤ 2 seconds
- Interruption response $\leq 300\text{ms}$
- 99.9% call-state consistency
- AI failure must not crash campaign workers

15. Out of Scope (Explicit)

- Voice cloning UI
 - Emotion synthesis tuning
 - External AI marketplace
-

Ai Calling

MODULE 4 — AI Calling Engine (Core Differentiator)

Version: 1.0

Status: Draft → Pending Confirmation

System: Omni-Channel CRM Platform

Audience: Product, Backend, AI/ML, Telephony, QA, Compliance

1. Module Overview & Strategic Intent

1.1 Purpose

This module is the **primary value creation engine** of the platform.

Everything else—CRM, campaigns, analytics, billing—exists to **enable, control, and monetize** this engine.

The AI Calling Engine must:

- Sound human
- Act intelligently
- Remain controllable
- Never lose cost discipline
- Stay compliant and auditable

This is **not**:

- A chatbot with a phone number
- A scripted IVR clone

This is a **stateful, goal-driven voice system**, deeply bound to CRM data and campaign logic.

2. Module Scope

2.1 In Scope

- Outbound AI calls
- Inbound AI calls
- Real-time conversation handling

- CRM updates via voice interactions
- Human handoff workflows
- Failure detection & recovery

2.2 Explicitly Out of Scope

- Telephony vendor management UI
 - Billing & invoicing logic (Module 9)
 - Post-call analytics dashboards (Module 7)
-

3. Call Types (Confirmed)

3.1 Supported Directions

- Outbound calls
- Inbound calls

Both must support:

- AI-first handling
- Human fallback
- CRM context injection from Module 3

No call exists without CRM context resolution.

4. AI Agent Model

4.1 AI Agent Types (Confirmed)

Supported agent archetypes include:

- Sales Agent
- Support Agent
- Collections Agent
- Survey Agent

Each AI agent type is:

- Configurable
- Reusable

- Deployable across multiple campaigns
-

4.2 AI Agent Identity

Each AI agent must have:

- Agent ID (system-generated)
- Agent type
- Supported language(s)
- Voice profile
- Personality tone
- Default objectives

Hard rule:

AI agents are **tenant-scoped**, not user-scoped.

This aligns with Module 2's AI vs User boundary.

5. Conversation Intelligence

5.1 Core Capabilities (Confirmed)

The AI Calling Engine must support:

- Natural turn-taking
- Interrupt handling
- Backchanneling (acknowledgments like “okay”, “I see”)
- Context carryover within the same call
- Intent detection
- Objection handling

No scripted IVR behavior unless explicitly configured.

5.2 Context Sources

AI must consume context from:

- Lead data (Module 3)
- Previous call summaries

- Campaign instructions
- Tenant-level rules
- Time, date, and timezone

Rule:

Context injection must be:

- Deterministic
 - Logged
 - Auditable
-

6. Conversation Design Model (Hybrid — Confirmed)

6.1 Prompt-Based Design

Includes:

- System prompt
- Goal prompt
- Guardrails
- Compliance instructions

Used for:

- Fast setup
 - Simple campaigns
 - Low-risk interactions
-

6.2 Flow-Based Design

Includes:

- Node-based conversation flows
- Conditional branching
- Data capture nodes
- Action execution nodes

Used for:

- Regulated workflows
- High-stakes calls

- Enterprise customers

Rule:

Prompt-based and flow-based approaches may coexist within the same tenant.

7. Call Lifecycle State Machine

Every call must follow a **strict, immutable state model**:

1. Initialized
2. Dialing / Ringing
3. Connected
4. Active Conversation
5. Action Execution
6. Wrap-up
7. Completed / Failed

All state transitions must be:

- Logged
 - Time-stamped
 - Non-editable
-

8. Actions Executable by AI

8.1 Allowed Actions

AI agents may:

- Update lead fields
 - Change lead stage
 - Create structured notes
 - Schedule callbacks
 - Trigger WhatsApp / SMS follow-ups
 - Transfer call to human
 - Intentionally end call
-

8.2 Forbidden Actions

AI agents may **not**:

- Modify billing or license data
- Delete CRM records
- Override compliance constraints

Enforcement must occur at API level.

9. Fallback & Failure Handling (Confirmed)

9.1 Failure Scenarios

The system must detect:

- ASR (speech-to-text) failure
 - LLM response failure
 - Tool / action execution failure
 - Silence or no response
 - User frustration or confusion signals
-

9.2 Fallback Actions

Based on campaign configuration, AI may:

- Retry (with configurable attempt limits)
- Transfer to human
- End call and tag lead
- Auto-schedule follow-up

Fallback behavior must be:

- Campaign-specific
 - Cost-aware
 - Fully logged
-

10. Human Handoff

10.1 Transfer Rules

Supported handoff types:

- Warm transfer (AI summarizes context)
- Cold transfer (optional, configurable)

System must:

- Check human agent availability
 - Prevent blind transfers
-

10.2 Post-Transfer Behavior

Upon transfer:

- AI billing stops immediately
 - Call ownership shifts to human
 - Conversation summary is auto-generated
 - Summary is stored in lead timeline
-

11. Multi-Language Support

- Language auto-detection (optional)
 - Language forced per campaign
 - Voice and prompt must remain consistent
 - Mid-call language change must be explicit and logged
-

12. Call Recording & Transcription

- Call recording enabled per tenant/campaign
 - Transcripts generated post-call
 - Stored against lead timeline (Module 3)
 - Consent capture required and logged
-

13. Cost Control & Safeguards (Non-Negotiable)

The system must enforce:

- Maximum call duration per campaign
- Silence timeout thresholds
- Repetitive loop detection
- Forced termination rules

Hard rule:

No AI call may run endlessly under any condition.

Cost discipline is a **platform responsibility**, not a tenant problem.

14. Security, Audit & Compliance

- Every AI action must be attributable
 - All call events must be logged
 - No cross-tenant call access
 - Compliance rules override campaign logic
-

V2.1

MODULE 4 — AI CALLING ENGINE

Document Type: Business Requirements + Functional Specification (BRD/FSD Hybrid)

Version: 2.1 — Detailed Draft

1. Executive Overview

1.1 Module Purpose

The AI Calling Engine serves as the **primary conversational execution layer** of the platform, enabling automated, AI-driven voice interactions across outbound and inbound communication channels.

The engine is designed as a **dual-deployment SaaS capability**:

- Operable as a standalone AI calling product
- Deeply integrated when deployed alongside the native CRM
 - Communication tool

This ensures adoption flexibility for clients with varying technology maturity.

1.2 Business Objectives

The module is intended to:

- Reduce human calling dependency
 - Scale outreach operations
 - Standardize conversation quality
 - Automate lead engagement workflows
 - Improve collections and support reachability
 - Lower cost-per-call operations
-

2. Deployment Models - Input

2.1 Standalone SaaS Deployment

In this model, the AI Calling Engine operates independently of the platform CRM.

Organizations may:

- Upload leads manually or via CSV
- Push leads via API – hitech erp
- Sync from external CRMs
- Receive call outcomes via webhook - output
- Hitech erp integration – Notification module

Functional Behavior

Lead data is stored within the calling module's scoped datastore. AI conversations reference only the attributes provided at ingestion or generated within call history.

No native pipeline or deal tracking exists unless synced externally.

2.2 Native CRM-Integrated Deployment

When integrated, the AI engine becomes a **conversation execution layer** on top of CRM data.

It can:

- Pull full lead profiles
- Reference interaction timelines
- Update deal/lead stages
- Trigger workflows
- Log transcripts automatically
- Task data automation
-

This enables deeper contextual conversations and lifecycle automation.

Campaign for AI calling only

Campaign to be plan - doc shared by hitech

3. Call Types (Confirmed)

3.1 Supported Directions

- Outbound calls
- Inbound calls

Both must support:

- AI-first handling
- Human fallback
- CRM context injection from Module 3

No call exists without CRM context resolution.

4. AI Agent Model

4.1 AI Agent Types (Confirmed)

Supported agent archetypes include:

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- Context carryover within the same call
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No scripted IVR behavior unless explicitly configured.

5.2 Context Sources

AI must consume context from:

- Lead data (Module 3)
- Previous call summaries
- Campaign instructions
- Tenant-level rules
- Time, date, and timezone

Rule:

Context injection must be:

- Deterministic

- Logged
 - Auditable
-

6. Conversation Design Model (Hybrid — Confirmed)

6.1 Prompt-Based Design

Includes:

- System prompt
- Goal prompt
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- Compliance instructions

Used for:

- Fast setup
 - Simple campaigns
 - Low-risk interactions
-

6.2 Flow-Based Design

Includes:

- Node-based conversation flows
- Conditional branching
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- Action execution nodes

Used for:

- Regulated workflows
- High-stakes calls
- Enterprise customers

Rule:

Prompt-based and flow-based approaches may coexist within the same tenant.

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 - Non-editable
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8. Actions Executable by AI

8.1 Allowed Actions

AI agents may:

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 - Change lead stage
 - Create structured notes
 - Schedule callbacks
 - Trigger WhatsApp / SMS follow-ups
 - Transfer call to human
 - Intentionally end call
-

8.2 Forbidden Actions

AI agents may **not**:

- Modify billing or license data
- Delete CRM records
- Override compliance constraints

Enforcement must occur at API level.

9. Fallback & Failure Handling (Confirmed)

9.1 Failure Scenarios

The system must detect:

- ASR (speech-to-text) failure
 - LLM response failure
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Based on campaign configuration, AI may:

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Fallback behavior must be:

- Campaign-specific
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 - Fully logged
-

10. Human Handoff

10.1 Transfer Rules

Supported handoff types:

- Warm transfer (AI summarizes context)
- Cold transfer (optional, configurable)

System must:

- Check human agent availability
 - Prevent blind transfers
-

10.2 Post-Transfer Behavior

Upon transfer:

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 - Call ownership shifts to human
 - Conversation summary is auto-generated
 - Summary is stored in lead timeline
-

11. Multi-Language Support

- Language auto-detection (optional)
 - Language forced per campaign
 - Voice and prompt must remain consistent
 - Mid-call language change must be explicit and logged
-

12. Call Recording & Transcription

- Call recording enabled per tenant/campaign
- Transcripts generated post-call
- Stored against lead timeline (Module 3)
- Consent capture required and logged

Whatsapp CRM

MODULE 4A — WhatsApp CRM & Messaging Engine

Parent Module: Module 4 — AI Calling Engine

SRS v1.0 | Status: Draft → Pending Confirmation

Category: Omni-Channel Engagement (Conversational CRM)

1. Purpose

This sub-module enables **WhatsApp as a fully integrated CRM communication channel**, tightly coupled with:

- Lead Management (Module 3)
- AI Calling Engine (Module 4)
- User & Role permissions (Module 2)
- Tenant & License enforcement (Module 1)

The objective is **not messaging**, but **conversation-driven conversion, support, and automation**.

WhatsApp becomes:

- A lead capture channel
 - A follow-up channel
 - A campaign execution channel
 - An AI + human hybrid engagement layer
-

2. Scope

2.1 In Scope

- Inbound & outbound WhatsApp messaging
- Lead creation & enrichment from WhatsApp
- AI-driven WhatsApp conversations
- Human-assisted WhatsApp handling
- Campaign-based WhatsApp messaging

- WhatsApp templates & approvals
- Media, documents, and interactive messages
- CRM timeline integration

2.2 Explicitly Out of Scope

- WhatsApp provider management UI
 - Direct WhatsApp billing reconciliation
 - External WhatsApp CRM sync
-

3. WhatsApp Channel Model

3.1 WhatsApp Account Model

- Platform-owned WhatsApp Business numbers
- Tenant-scoped WhatsApp identity
- No BYO WhatsApp credentials (aligned with telephony rules)

Each tenant may have:

- One or more WhatsApp sender identities (configurable by license)
-

3.2 Conversation Ownership

Each WhatsApp conversation must be linked to:

- Tenant ID
 - Lead ID (existing or auto-created)
 - Assigned owner (user or AI)
 - Campaign ID (if applicable)
-

4. WhatsApp Lead Capture & CRM Binding

4.1 Inbound Message Handling

On first inbound WhatsApp message:

- System checks for existing lead via phone number

- If not found → auto-create lead
- Source = WhatsApp
- First-touch timestamp recorded

Captured metadata:

- WhatsApp number
 - Display name
 - Message content
 - Media type (if any)
 - Timestamp
-

4.2 Lead Enrichment via WhatsApp

AI or human can:

- Ask qualification questions
- Update lead fields
- Apply tags
- Move lead stages
- Assign pipelines

All updates follow **Module 3 lead rules**.

5. Messaging Capabilities

5.1 Message Types Supported

- Text
 - Image
 - Video
 - Audio
 - Documents
 - Stickers (optional)
 - Location
 - Contact cards
-

5.2 Interactive Messages

- Quick reply buttons
- List messages
- CTA buttons (call, link)
- Template-driven structured messages

User responses must be mapped back to:

- Lead fields
 - Stage changes
 - AI intents
-

6. AI-Powered WhatsApp Conversations

6.1 AI Capabilities

AI is allowed to:

- Reply to inbound messages
- Ask structured questions
- Handle FAQs
- Qualify leads
- Trigger campaigns
- Schedule callbacks
- Escalate to human

AI is **not allowed** to:

- Send unapproved templates
 - Modify billing or compliance fields
 - Spam or loop messages
-

6.2 AI Context Sources

AI must consume context from:

- Lead data (Module 3)
- Past WhatsApp messages
- Call summaries
- Campaign instructions
- Tenant rules & compliance constraints

Context injection must be logged and auditable.

7. **Human + AI Handoff Model**

7.1 Ownership Modes

- AI-owned conversation
- Human-owned conversation
- Hybrid (AI assist, human sends)

Switching ownership must:

- Be explicit
 - Be logged
 - Update SLA timers
-

7.2 Supervisor Controls

Managers can:

- Take over conversations
 - Reassign ownership
 - Freeze AI replies
 - Audit message history
-

8. **WhatsApp Campaigns**

8.1 Campaign Types

- Broadcast campaigns
 - Drip sequences
 - Event-based triggers
 - AI-initiated follow-ups
 - Call-to-WhatsApp campaigns
-

8.2 Campaign Rules

- Template-only initiation (compliance)
- Opt-in enforcement
- Rate limiting
- Time-window enforcement
- Cost-aware execution

Campaign outcomes must update:

- Lead stage
 - Lead score
 - Activity timeline
-

9. **Template Management**

9.1 Template Lifecycle

- Draft
- Submitted
- Approved
- Rejected
- Archived

Templates include:

- Language
 - Category (utility, marketing, auth)
 - Variables
 - CTA buttons
-

9.2 Usage Rules

- Only approved templates can initiate conversations
 - Free-text allowed only within active session window
 - Template usage logged per campaign and lead
-

10. **Automation & Workflow Triggers**

WhatsApp events can trigger:

- Lead stage change
- Task creation
- Callback scheduling
- AI call initiation
- Campaign enrollment
- SLA escalation

Triggers must be configurable and auditable.

11. Activity Timeline Integration

Every WhatsApp event must appear in the **Lead Timeline (Module 3)**:

- Messages sent
- Messages received
- Media shared
- AI actions
- Human actions
- Campaign touches

Timeline is:

- Read-only
 - Chronological
 - Immutable
-

12. Search, Filters & Views

Users can filter WhatsApp conversations by:

- Lead stage
- Assigned owner
- Unread status
- SLA breach
- Campaign
- Tags
- AI vs human handled

Saved views supported per user and role.

13. **Permissions & Security**

13.1 Role-Based Controls

Permissions include:

- View conversations
- Reply manually
- Approve templates
- Enable/disable AI replies
- Export conversations (if allowed)

All permissions enforced via **Module 2 RBAC**.

13.2 Compliance & Privacy

- Consent tracking required
 - Opt-out respected immediately
 - Message deletion rules enforced
 - Data retention aligned with tenant policy
-

14. **Cost Control & Safeguards**

- Message rate limiting
- Daily send caps per campaign
- AI reply throttling
- Infinite loop detection
- Forced AI pause rules

No uncontrolled WhatsApp spend allowed.

15. Reporting Hooks (Data Only)

This module will emit data for:

- Message delivery status
- Response rates
- Conversion events

- AI vs human performance
- Campaign effectiveness

Dashboards are handled in **Module 7**.

whatsapp CRM V2.2

WhatsApp CRM & Messaging Engine

Standalone SaaS + Optional CRM-Integrated Module

Version 1.0

Status: Draft

1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the WhatsApp CRM & Messaging Engine.

The module shall operate in two modes:

1. Standalone WhatsApp Engagement Platform
2. Integrated Module within the Planned SaaS CRM Suite

The system enables WhatsApp as a structured, compliant, automation-ready conversational channel for lead generation, qualification, campaign execution, support, and revenue acceleration.

This is not a messaging tool.

It is a conversion engine with auditability, AI governance, and role-based control.

1.2 Product Positioning

The module supports:

- Lead Capture via WhatsApp
- AI + Human Hybrid Conversations
- Campaign Orchestration
- Workflow Automation
- Compliance Enforcement
- Performance Reporting

When integrated with CRM:

- Leads sync to CRM
 - Conversations attach to CRM timeline
 - Pipeline stages update automatically
 - Lead scoring integrates
 - Deal tracking supported
-

2. Scope

2.1 In Scope

- Inbound WhatsApp Messaging
 - Outbound Messaging (session + template)
 - Lead Auto-Creation
 - Campaign Broadcasting
 - AI-driven Conversations
 - Human Escalation
 - Automation Triggers
 - Role-Based Access
 - Consent Tracking
 - Reporting
 - Public API & Webhooks
-

3. System Modes

3.1 Standalone Mode

System maintains:

- Internal Lead Database
- Internal Timeline
- Internal Pipeline
- Internal User Roles
- Internal Campaign Engine

3.2 CRM-Integrated Mode

If CRM subscription active:

- Leads bind to CRM module
 - Timeline writes to CRM Lead Activity Log
 - Stage changes sync
 - AI context pulls CRM data
 - Campaign outcomes affect CRM scoring
-

4. WhatsApp Account & Tenant Model

- Tenant-scoped identity through meta
- BYO credentials/ apply through platform

Meta verification outside the system

Our system will give access to submit information and rest meta will take care

Each conversation must bind to:

- Tenant ID
 - Lead ID
 - Assigned Owner
 - Campaign ID (optional)
 - Conversation Ownership Type (AI/Human/Hybrid)
-

5. Lead Capture & Binding

5.1 Inbound Lead Creation

When a new WhatsApp message is received:

System shall:

- Match by phone number

- If no match → auto-create lead
- Capture metadata
- Tag source = WhatsApp
- Record first-touch timestamp

User Stories

As an Agent

I want incoming WhatsApp messages to auto-create leads
So that no inquiry is missed.

As a Manager

I want lead source tagging automated
So that channel attribution is accurate.

5.2 Lead Enrichment

System shall allow:

- AI qualification questions
- Field updates via conversation
- Tag application
- Stage updates
- Pipeline movement

In CRM mode:

Changes must reflect in CRM Lead Object.

User Stories

As an Agent

I want to update lead stage directly from chat
So that workflow remains centralized.

As a Manager

I want AI to auto-qualify using structured logic
So that high-intent leads are prioritized.

6. Messaging Capabilities

6.1 Supported Message Types

- Text
- Image
- Video
- Audio
- Documents
- Location
- Interactive Buttons
- List Messages
- CTA Buttons
- Template Messages

6.2 Messaging Rules

- **Free-text allowed only within 24-hour session window :**
<https://business.whatsapp.com/products/platform-pricing>
- Template required to initiate conversation
- Template approval lifecycle enforced
- Rate limits applied per tenant

User Stories

As an Agent

I want to send interactive CTAs

So that customers can take direct action.

As an Admin

I want template approval workflow enforced

So that compliance risk is minimized.

7. AI Conversation Engine

7.1 AI Capabilities

AI may:

- Reply to inbound messages
- Ask structured qualification questions

- Handle FAQs
- Schedule callbacks
- Trigger campaigns
- Escalate to human
- Update lead fields

AI may NOT:

- Send unapproved templates
- Modify billing data
- Override compliance restrictions
- Loop infinite responses

7.2 AI Context Injection

AI context sources:

- Lead data
- Past WhatsApp messages
- Call summaries
- Campaign objectives
- Tenant compliance rules

All AI context injection must be logged and auditable.

User Stories

As a Manager

I want AI responses governed by compliance rules
So that brand risk is controlled.

As an Agent

I want AI suggestions visible but editable
So that control remains human-led.

8. Human + AI Handoff Model

8.1 Ownership Modes

- AI-Owned

- Human-Owned
- Hybrid Assist

Switching ownership must:

- Be explicit
- Be logged
- Reset SLA timers

8.2 Escalation Triggers

- Customer request
- AI low-confidence score
- Negative sentiment detection
- Compliance-sensitive query

User Stories

As an Agent

I want full transcript visible when AI escalates
So that context is preserved.

As a Supervisor

I want ability to freeze AI replies
So that escalation can be controlled.

9. Campaign Engine

9.1 Campaign Types

- Broadcast → verified broadcast list
- Drip Sequence
- Event-Based Trigger
- AI Follow-up Campaign
- Call-to-WhatsApp

9.2 Campaign Rules

- Template-only initiation

- Opt-in enforcement
- Daily send caps
- Time window restrictions
- Budget guardrails

Campaign outcomes must update:

Standalone Mode:

- Lead Stage
- Lead Score
- Timeline Entry

CRM Mode:

- CRM Lead Stage
- CRM Score
- CRM Deal Object (if exists)

User Stories

As a Manager

I want broadcast campaigns limited by opt-in rules
So that compliance is enforced automatically.

As a Revenue Lead

I want drip campaigns tied to lead stage
So that nurturing aligns with pipeline position.

10. Template Management

Lifecycle:

- Draft
- Submitted
- Approved
- Rejected
- Archived

Template attributes:

- Language

- Category
- Variables
- CTA buttons
- Versioning

All template usage logged per:

- Lead
 - Campaign
 - Tenant
-

11. Automation & Workflow Triggers

WhatsApp events may trigger:

- Lead Stage Change
- Task Creation
- Callback Scheduling
- AI Call Initiation
- Campaign Enrollment
- SLA Escalation

Triggers must be:

- Configurable
- Role-restricted
- Auditable

User Stories

As an Operations Head
I want WhatsApp replies to trigger pipeline progression
So that system operates event-driven.

12. Activity Timeline

Every WhatsApp event must appear in timeline:

- Sent messages
- Received messages
- AI responses
- Human responses
- Media shared
- Campaign touches
- Stage changes

Timeline is:

- Chronological
- Immutable
- Read-only

CRM Mode:

Timeline syncs to CRM Activity Log.

13. Search & Filters

Filters:

- Lead Stage
- Owner
- Unread Status
- SLA Breach
- Campaign
- Tags
- AI vs Human handled

Saved Views supported per user role.

14. Permissions & Security

14.1 Role-Based Permissions

Permissions include:

- View Conversations

- Reply Manually
- Approve Templates
- Enable/Disable AI
- Export Conversations
- Create Campaigns
- Configure Automations

All enforced via RBAC engine.

14.2 Compliance & Privacy

System must:

- Track consent timestamp
 - Enforce opt-out immediately
 - Block messaging to non-consented users
 - Respect retention policy per tenant
 - Support data deletion request
-

15. Cost Control & Safeguards

System must include:

- Rate limiting
- Daily caps per campaign
- AI throttling
- Infinite loop detection
- Forced AI pause logic
- Tenant-level spending guardrails

No uncontrolled WhatsApp spend allowed.

16. Reporting & Analytics

System emits:

- Message Delivery Status
- Read Rates

- Response Rates
- Conversion Events
- AI vs Human Performance
- Campaign Effectiveness
- Lead Velocity Metrics
- Revenue Attribution (CRM Mode)

Dashboards available in Reporting Module.

17. API & Third-Party Integration

System shall expose:

- **REST API**
- **Webhooks**
- Auth via OAuth/API Keys
- Versioned endpoints

Supported external integration:

- Third-party CRM sync
- ERP systems
- Marketing automation tools
- Custom analytics

Events exposed:

- New Lead
 - New Message
 - Campaign Sent
 - Stage Updated
 - Opt-In Changed
 - SLA Breach
-

18. Non-Functional Requirements

- 99.9% uptime
- Horizontal scalability

- Secure multi-tenant isolation
 - Encryption in transit & at rest
 - Audit logs retained minimum 12 months
 - Sub-second UI response time
 - AI guardrails enforced at runtime
-

19. Future Enhancements

- Multi-language AI routing
 - WhatsApp Commerce Catalog integration
 - Revenue prediction modeling
 - AI intent clustering
 - Cross-channel orchestration engine
-

1. Whatsapp workflow data collection → campaign → templates based and promotional
2. Centralised system : payment, reports, → transactional
3. Communication on CRM at one place
4. Whatsapp catalog
5. Whatsapp communication
6. Whatsapp AI agent
7. Whatsapp business is must

Meta Business api

Business number

Subscription : info

Templates bante ho

Approval from meta

App wo templates send kr skte ho in a campaign

Agar user ne 24 hours me reply kiya then you can send message

Template followup

- direc

SMS

MODULE 4B — SMS CRM & Messaging Engine

Parent Module: Module 4 — AI Calling Engine

SRS v1.0 | Status: Draft → Pending Confirmation

Category: Omni-Channel Engagement (Conversational CRM)

1. Purpose

This sub-module enables **SMS as a first-class CRM communication channel**, tightly integrated with:

- Lead Management (Module 3)
- AI Calling Engine (Module 4)
- WhatsApp CRM (Module 4A)
- User & Role Management (Module 2)
- Tenant & License enforcement (Module 1)

SMS is positioned as:

- A high-deliverability follow-up channel
 - A fallback when rich channels fail
 - A transactional & alert medium
 - A cost-predictable engagement layer
-

2. Scope

2.1 In Scope

- Inbound & outbound SMS
- Lead creation & enrichment from SMS
- AI-assisted SMS conversations
- Human-driven SMS handling
- Campaign-based SMS messaging
- Template management
- Short-link & tracking support

- CRM timeline integration

2.2 Explicitly Out of Scope

- SMS provider management UI
 - Direct SMS carrier billing reconciliation
 - External SMS CRM synchronization
-

3. SMS Channel Model

3.1 SMS Sender Identity

- Platform-owned sender IDs / long codes / short codes
- Tenant-scoped sender allocation
- No BYO SMS gateways (aligned with telephony & WhatsApp rules)

Sender configuration governed by license & region.

3.2 Conversation Model

Each SMS thread must be associated with:

- Tenant ID
- Lead ID (existing or auto-created)
- Assigned owner (user or AI)
- Campaign ID (if applicable)

Threads are **stateful**, not stateless blasts.

4. SMS Lead Capture & CRM Binding

4.1 Inbound SMS Handling

On first inbound SMS:

- Match lead via phone number
- If not found → auto-create lead
- Source = SMS

- First-touch timestamp recorded

Captured metadata:

- Sender number
 - Message content
 - Timestamp
 - Keyword (if applicable)
-

4.2 Keyword-Based Lead Actions

SMS keywords can trigger:

- Lead creation
- Pipeline entry
- Campaign enrollment
- Opt-in / opt-out
- AI call initiation

Keywords are tenant-configurable.

5. Messaging Capabilities

5.1 Message Types Supported

- Plain text
- Unicode / regional language text
- Short links (tracked)
- Opt-in / opt-out messages

Media (MMS) support is optional and license-controlled.

5.2 Personalization Tokens

SMS templates may include:

- Lead name
- Company
- Appointment time

- Agent name
- Dynamic short links

Tokens must be validated before send.

6. AI-Powered SMS Conversations

6.1 AI Capabilities

AI is allowed to:

- Respond to inbound SMS
- Ask short qualification questions
- Confirm appointments
- Share links
- Escalate to human
- Trigger AI calls or WhatsApp follow-ups

AI is **not allowed** to:

- Send unsolicited messages
 - Bypass opt-out rules
 - Modify compliance-sensitive fields
-

6.2 AI Context Sources

AI consumes context from:

- Lead profile
- Previous SMS threads
- Call summaries
- Campaign goals
- Tenant compliance rules

Context injection must be deterministic and logged.

7. Human + AI Collaboration Model

7.1 Ownership Modes

- AI-handled SMS
- Human-handled SMS
- Hybrid (AI drafts, human sends)

Ownership transitions must be explicit and logged.

7.2 Supervisor Controls

Managers may:

- Take over threads
 - Pause AI replies
 - Reassign ownership
 - Audit conversation history
-

8. SMS Campaigns

8.1 Campaign Types

- Broadcast SMS
 - Drip sequences
 - Event-triggered messages
 - Reminder & alert campaigns
 - Call-to-SMS campaigns
-

8.2 Campaign Controls

- Opt-in enforcement
- Rate limiting
- Quiet hours / DND windows
- Per-campaign budget caps
- Cost-aware throttling

Campaign execution must update:

- Lead timeline
 - Lead stage
 - Engagement score
-

9. Template Management

9.1 Template Lifecycle

- Draft
- Approved
- Active
- Archived

Templates include:

- Message body
 - Language
 - Category (transactional, marketing, alert)
 - Compliance tags
-

9.2 Usage Rules

- Templates required for bulk sends
 - Free-text allowed only in active conversations
 - All sends logged per lead & campaign
-

10. Automation & Workflow Triggers

SMS events may trigger:

- Lead stage transitions
- Task creation
- Callback scheduling
- AI call initiation
- SLA escalation
- Campaign branching

All automations must be auditable.

11. Activity Timeline Integration

Every SMS event must appear in the **Lead Timeline (Module 3)**:

- Messages sent
- Messages received
- AI vs human actions
- Campaign interactions
- Keyword triggers

Timeline is immutable and chronological.

12. Search, Filters & Views

Users can filter SMS conversations by:

- Lead stage
- Assigned owner
- Unread status
- Campaign
- Keyword
- Opt-out status

Saved views supported.

13. Permissions & Security

13.1 Role-Based Permissions

Permissions include:

- View SMS threads
- Send messages
- Approve templates
- Export conversations (if allowed)
- Disable AI responses

Controlled via **Module 2 RBAC**.

13.2 Compliance & Privacy

- Mandatory opt-out handling (STOP, UNSUBSCRIBE)
 - Opt-out applied immediately across all campaigns
 - Message retention policies enforced
 - Region-specific compliance supported
-

14. Cost Control & Safeguards

- Per-campaign SMS caps
- Daily send limits
- AI reply throttling
- Loop detection
- Forced shutdown rules

No uncontrolled SMS spend permitted.

15. Reporting Hooks (Data Emission Only)

SMS module must emit:

- Delivery status
- Response rate
- Keyword engagement
- Conversion triggers
- AI vs human effectiveness

Visualization handled in **Module 7**.

Questions regarding compliances

Email

MODULE 4C — Email CRM & Communication Engine

Parent Module: Module 4 — AI Calling Engine

SRS v1.0 | Status: Draft → Pending Confirmation

Category: Omni-Channel Engagement (Conversational CRM)

1. Purpose

This sub-module enables **Email as a first-class CRM engagement channel**, tightly integrated with:

- Lead Management (Module 3)
- AI Calling Engine (Module 4)
- WhatsApp CRM (Module 4A)
- SMS CRM (Module 4B)
- User & Role Management (Module 2)
- Tenant & License Management (Module 1)

Email is positioned as:

- A high-context, long-form communication channel
 - A campaign and nurture backbone
 - A compliance-friendly audit medium
 - A bridge between automation and human interaction
-

2. Scope

2.1 In Scope

- Inbound & outbound email handling
- Lead creation & enrichment from email
- AI-assisted email drafting & replies
- Human-driven email conversations
- Campaign-based email sending
- Template & signature management
- Threading & reply tracking
- CRM timeline integration
- Email server / provider configuration UI

- Email deliverability analytics (SPF/DKIM dashboards)
 - External email CRM synchronization
-

3. Email Channel Model

3.1 Sender Identity

- Platform-managed sending domains
- Tenant-scoped sender addresses (e.g., support@, sales@)
- Optional user-based sender aliases (license-controlled)
- No BYO SMTP credentials in v1

All sender identities are:

- Verified
 - Audited
 - License-governed
-

3.2 Conversation & Thread Model

Each email thread must be associated with:

- Tenant ID
- Lead ID (existing or auto-created)
- Assigned owner (user or AI)
- Campaign ID (if applicable)
- Thread ID (immutable)

Email threads are treated as **stateful conversations**, not isolated messages.

4. Email Lead Capture & CRM Binding

4.1 Inbound Email Handling

On inbound email:

- Match lead via sender email
- If not found → auto-create lead

- Source = Email
- First-touch timestamp recorded

Captured metadata:

- Sender email
 - Subject
 - Body (HTML + text)
 - Attachments
 - Timestamp
-

4.2 Email-Based Lead Actions

Emails may trigger:

- Lead stage updates
- Task creation
- Callback scheduling
- Campaign enrollment
- AI follow-up calls

Triggers must be rule-based and auditable.

5. Email Messaging Capabilities

5.1 Supported Content

- Plain text
 - HTML emails
 - Inline images
 - File attachments
 - Tracking links
 - Email signatures
-

5.2 Personalization Tokens

Templates may include:

- Lead name

- Company name
- Assigned agent
- Appointment details
- Dynamic CTA links

Token validation required before send.

6. AI-Powered Email Assistance

6.1 AI Capabilities

AI is allowed to:

- Draft outbound emails
- Suggest replies
- Summarize inbound threads
- Tag sentiment & intent
- Recommend next actions
- Trigger cross-channel follow-ups

AI is **not allowed** to:

- Send bulk emails without approval
 - Bypass opt-out rules
 - Modify compliance-sensitive fields
-

6.2 AI Context Sources

AI consumes:

- Lead profile & history
- Past emails & threads
- Call transcripts
- WhatsApp/SMS interactions
- Campaign objectives
- Tenant compliance rules

Context injection must be logged.

7. Human + AI Collaboration Model

7.1 Ownership Modes

- AI-drafted, human-approved
- Human-authored
- Fully AI-managed (transactional use cases)

Ownership must be explicit per message.

7.2 Supervisor Controls

Managers may:

- Review drafts
 - Enforce approvals
 - Reassign threads
 - Freeze AI sending
-

8. Email Campaigns

8.1 Campaign Types

- One-time broadcasts
 - Drip/nurture campaigns
 - Event-triggered emails
 - Transactional notifications
 - Call-to-email campaigns
-

8.2 Campaign Controls

- Template approval enforcement
- Opt-in & unsubscribe handling
- Quiet hours (configurable)
- Per-campaign send caps
- Cost-aware throttling

Campaign interactions must update:

- Lead timeline
 - Engagement score
 - Lead stage (where applicable)
-

9. Template & Signature Management

9.1 Template Lifecycle

- Draft
- Approved
- Active
- Archived

Templates include:

- Subject
 - Body (HTML/text)
 - Variables
 - Language
 - Category (marketing, transactional, support)
-

9.2 Signatures

- Tenant-level signatures
 - Role-based signatures
 - User-specific signatures (optional)
-

10. Automation & Workflow Triggers

Email events may trigger:

- Lead stage changes
- Task creation
- SLA escalation
- AI call initiation
- WhatsApp/SMS fallback
- Campaign branching

Automations must be deterministic and logged.

11. Activity Timeline Integration

Every email event must appear in the **Lead Timeline (Module 3)**:

- Sent emails
- Received emails
- Attachments
- AI actions
- Human actions
- Campaign interactions

Timeline is immutable and chronological.

12. Search, Filters & Views

Users can search/filter by:

- Subject
- Sender / recipient
- Lead stage
- Campaign
- Unread status
- Sentiment (AI-tagged)
- Attachment presence

Saved views supported.

13. Permissions & Security

13.1 Role-Based Permissions

Permissions include:

- View email threads
- Draft emails
- Send emails

- Approve templates
- Export emails (if allowed)

Controlled **via Module 12 RBAC.**

13.2 Compliance & Privacy

- Unsubscribe handling mandatory
 - Email suppression lists enforced
 - Data retention policies applied
 - Attachment scanning supported (optional)
-

14. Cost Control & Safeguards

- Daily send limits
- Per-campaign caps
- AI draft throttling
- Spam-risk prevention rules
- Forced suspension on abuse detection

No uncontrolled email volume allowed.

15. Reporting Hooks (Data Emission Only)

Email module emits:

- Delivery & open events
- Click-through actions
- Reply rates
- Conversion triggers
- AI vs human performance

Dashboards handled in **Module 7.**

MODULE 5 — Campaigns & Workflow Automation

MODULE 5 — Campaigns & Workflow Automation

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module determines **when, why, and how** the AI Calling Engine engages leads. It is the **orchestration layer** that converts static CRM data into dynamic revenue-driving motion.

Think of it as the **brainstem** between CRM (Module 3) and AI Calling (Module 4).

Poor automation = wasted minutes.

This module is designed for **precision, control, and cost efficiency**.

2. Scope

Applies to:

- Campaign creation & management
- Event-driven and batch workflows
- AI call scheduling & retries
- Multi-channel follow-ups
- Visual workflow orchestration

Does **not** include:

- Call execution logic (Module 4)
 - Revenue tracking (Module 6)
 - Analytics dashboards (Module 7)
-

3. Campaign Types (Confirmed)

3.1 Supported Campaign Modes

- **Event-driven campaigns**
 - Triggered by:
 - New lead creation
 - Stage change
 - Score threshold
 - Inactivity timeout
- **Batch campaigns**
 - Scheduled runs
 - Lead list-based
 - Time-window constrained

Both modes can coexist per tenant.

4. Campaign Entity Definition

Each campaign has:

- Campaign ID
- Name
- Campaign type (event / batch)
- Associated AI agent
- Target pipeline & stages
- Status (draft / active / paused / archived)
- Schedule & time windows
- Retry policy
- Cost safeguards

Campaigns are **tenant-scoped** and reusable.

5. Workflow Builder (Confirmed: Visual, Drag & Drop)

5.1 Workflow Design Model

- Node-based visual builder
 - Linear + conditional paths
 - Explicit start & end nodes
-

5.2 Supported Node Types

- Trigger nodes
 - Lead created
 - Stage changed
 - Time-based trigger
 - Action nodes
 - AI call
 - WhatsApp message
 - SMS
 - Task creation
 - Lead update
 - Decision nodes
 - Call outcome
 - Lead score check
 - Field condition
 - Delay nodes
 - Fixed delay
 - Business-hour aware delay
-

6. Campaign Scheduling & Time Rules

6.1 Scheduling Controls

- Start date & end date
 - Business hours
 - Timezone awareness (tenant + lead)
 - Holiday exclusions
-

6.2 Call Throttling

- Max calls per hour
- Max concurrent calls
- Per-campaign call caps

Hard limits enforced to prevent cost overruns.

7. Retry & Escalation Logic

7.1 Retry Rules (Confirmed)

- Retry on:
 - No answer
 - Busy
 - Call drop
 - Retry parameters:
 - Max attempts
 - Delay between retries
 - Channel fallback (WhatsApp/SMS)
-

7.2 Escalation Paths

- Transfer to human
 - Assign task to user
 - Move lead to escalation stage
 - Stop automation entirely
-

8. Multi-Channel Follow-Ups (Confirmed in v1)

Supported channels:

- WhatsApp
- SMS

Rules:

- Triggered by AI outcome
- Template-based
- Logged in lead timeline
- Opt-out compliant

Email is **explicitly excluded** from v1.

9. A/B Testing (Controlled)

9.1 Testable Variables

- AI prompt variants
 - Opening script
 - Call timing windows
-

9.2 Constraints

- One active test per campaign
- Traffic split configurable
- Cost impact visible before activation

No black-box experimentation.

10. Campaign State Management

Campaign states:

- Draft
- Active
- Paused
- Completed
- Archived

State transitions:

- Logged
- Permission-controlled
- Immediately enforced

Pausing a campaign:

- Stops new calls
 - Does not interrupt active calls
-

11. AI ↔ Workflow Boundary Rules

- AI executes **only what workflow allows**
- AI cannot:
 - Self-trigger campaigns
 - Override retry limits

- Bypass throttles

All autonomy is **bounded**.

12. Error Handling & Recovery

- Node-level failure isolation
- Retry on transient failures
- Campaign auto-pauses on systemic errors
- Admin alerts on abnormal behavior

No silent failures.

13. Non-Functional Requirements (Module-Specific)

- Workflow execution latency $\leq 500\text{ms}$
 - Visual builder must support:
 - ≥ 500 nodes per workflow
 - Campaign edits must not break active executions
 - Deterministic replay for debugging
-

14. Out of Scope (Explicit)

- Email automation
 - External workflow engines
 - Customer-written scripts/code
-

Module 5 revised

MODULE 5 — Campaign Management & Orchestration (Unified Omni-Channel Brain)

Version: 1.0

Status: Draft → Pending Confirmation

System: Omni-Channel Martech CRM Platform

Audience: Product, Backend, AI, Growth, QA, Compliance

1. Module Overview & Strategic Role

1.1 Purpose

This module is the **central orchestration engine** that coordinates **who to contact, when, through which channel, with what logic, and at what cost.**

If:

- Module 3 = data truth
- Module 4 = execution power
- Module 4A/B/C = channels

Then **Module 5 is the brain.**

It ensures:

- Consistent customer experience
 - Cost-disciplined execution
 - Cross-channel intelligence
 - Measurable outcomes
-

2. Scope

2.1 In Scope

- Campaign creation & configuration
- Omni-channel orchestration (Call, WhatsApp, SMS, Email)
- Audience segmentation
- Trigger-based and scheduled execution

- AI-driven branching logic
- Budget & usage control
- Campaign lifecycle management

2.2 Explicitly Out of Scope

- Channel-specific UI (handled in 4A/4B/4C)
 - Billing & invoicing (Module 9)
 - Analytics dashboards (Module 7)
-

3. Campaign Entity Definition

3.1 Core Campaign Attributes

Each campaign must include:

- Campaign ID (system generated)
 - Campaign Name
 - Campaign Type
 - Primary Channel
 - Secondary / Fallback Channels
 - Objective (configurable)
 - Status (Draft / Active / Paused / Completed / Archived)
 - Created by (User / System)
 - Start date & time
 - End date (optional)
-

3.2 Campaign Types (Confirmed)

Supported campaign archetypes:

- Sales outreach
- Lead qualification
- Follow-up / nurture
- Support notification
- Collections
- Surveys / feedback
- Transactional alerts

Campaign types influence:

- Allowed channels
 - AI behavior
 - Compliance rules
 - Cost limits
-

4. Audience Selection & Segmentation

4.1 Audience Definition

Campaign audience is defined via **CRM filters (Module 3)**:

- Lead source
- Pipeline & stage
- Score range
- Last activity
- Tags
- Assigned owner
- Consent status

Audiences are **dynamic by default** unless explicitly frozen.

4.2 Exclusion Rules

Campaigns must support exclusions:

- Opt-out leads
- DND / quiet hour conflicts
- Recently contacted leads
- Active conversations
- Manual suppression lists

Exclusions are evaluated **at execution time**, not only at campaign start.

5. Omni-Channel Execution Model

5.1 Channel Sequencing

Campaigns may define:

- Single-channel execution
- Multi-step channel sequences

Example:

1. AI Call
 2. If no answer → WhatsApp after 2 hours
 3. If no response → SMS next day
 4. If engaged → Email follow-up
-

5.2 Channel Selection Rules

Channel choice may depend on:

- Lead preference
- Language
- Timezone
- Cost thresholds
- Previous engagement history

Rules must be deterministic and logged.

6. AI-Driven Campaign Logic

6.1 AI Decision Capabilities

AI may:

- Decide next best channel
- Choose message variant
- Adjust timing
- Escalate to human
- Terminate campaign journey early

AI **cannot**:

- Override compliance rules
 - Exceed campaign budget
 - Ignore opt-out signals
-

6.2 AI Inputs

AI decisions are based on:

- Lead profile & history
 - Previous campaign outcomes
 - Channel performance
 - Time & context
 - Tenant-defined guardrails
-

7. Campaign Flow Designer (Hybrid)

7.1 Rule-Based Configuration

Supports:

- IF / ELSE conditions
- Delays
- Counters
- Retry limits

Used for:

- Simple campaigns
 - Predictable flows
-

7.2 Flow-Based Orchestration

Node-based flow builder with:

- Channel nodes (Call / WhatsApp / SMS / Email)
- Decision nodes
- Wait timers
- Exit conditions
- Error handling paths

Used for:

- Complex journeys
- Enterprise automation
- Compliance-heavy workflows

8. Execution Triggers

Campaigns may start based on:

- Manual trigger
- Scheduled time
- Lead creation
- Stage change
- AI call outcome
- Inactivity timeout
- External webhook (future)

Triggers must be logged with source attribution.

9. Cost Control & Budget Management (Critical)

9.1 Budget Definitions

Campaigns must support:

- Total budget cap
 - Daily spend limit
 - Per-lead spend limit
 - Channel-specific caps
-

9.2 Enforcement Rules

When limits are reached:

- Campaign auto-pauses
- Pending actions cancelled
- Admin notified

No soft overages allowed unless explicitly approved.

10. Campaign State Management

10.1 Campaign States

- Draft
- Active
- Paused
- Completed
- Archived

State transitions must be:

- Logged
 - Attributable
 - Non-destructive
-

10.2 Lead-Level Campaign State

Each lead maintains:

- Current campaign step
 - Last action
 - Next scheduled action
 - Exit reason (if exited)
-

11. Human Intervention & Overrides

11.1 Allowed Overrides

Authorized users may:

- Pause campaign
- Skip lead step
- Reassign lead
- Force channel change
- Escalate to human

Overrides must:

- Require permission
- Be logged
- Not break compliance rules

12. Failure Handling & Recovery

12.1 Failure Scenarios

- Channel delivery failure
- AI execution failure
- Time window violation
- Budget exhaustion

12.2 Recovery Actions

- Retry (configurable)
- Switch channel
- Exit campaign
- Schedule follow-up
- Flag for review

Failure logic must be campaign-specific.

13. Activity Timeline & Audit Integration

Every campaign action must be written to:

- Lead Timeline (Module 3)
- Campaign execution log
- AI decision log

Logs are:

- Immutable
 - Time-stamped
 - Filterable
-

14. Permissions & Security

- Campaign creation restricted by role
 - Execution permissions configurable
 - No cross-tenant campaign visibility
 - RBAC enforced via Module 2
 - License limits enforced via Module 1
-

15. Data Outputs (For Analytics Module)

This module emits:

- Campaign performance events
- Channel conversion markers
- Cost consumption data
- AI vs human effectiveness signals

Visualization is handled in **Module 7**.

Closing Position (Straight Talk)

This module determines **whether the platform scales or collapses**.

Without orchestration:

- Channels spam
- Costs leak
- AI behaves randomly

With orchestration:

- Conversations feel intentional
 - Spend stays controlled
 - AI becomes predictable, not scary
-

MODULE 6 — Deal & Pipeline Management

MODULE 6 — Deal & Pipeline Management

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module governs **post-qualification monetization**.

Once a lead is warm, this system ensures:

- Clear ownership
- Predictable progression
- Zero ambiguity on “what happens next”

It converts **conversation intent into accountable execution**.

2. Scope

Applies to:

- Opportunity creation
- Pipeline progression
- Deal ownership
- AI-assisted deal actions
- Win / loss capture

Does **not** include:

- Invoicing
 - Revenue forecasting (explicitly excluded)
 - Payment collection
-

3. Deal Entity Definition

3.1 Core Deal Attributes

Mandatory:

- Deal ID
- Associated Lead ID
- Pipeline ID
- Stage
- Owner (user)
- Created timestamp

Optional:

- Deal value
- Expected close date
- Priority
- Tags
- Custom fields (tenant-defined)

Deals are **child entities of leads**.

4. Pipelines (Confirmed: Multiple per Tenant)

4.1 Pipeline Structure

Each tenant can define:

- Multiple pipelines
 - Pipeline purpose (Sales, Renewals, Upsell, etc.)
 - Default entry stage
-

4.2 Pipeline Stages

- Fully configurable
- Ordered
- Stage-specific rules supported

Examples:

- Qualification
- Demo Scheduled
- Proposal Sent
- Negotiation

- Won
 - Lost
-

5. Deal Creation Rules

Deals can be created via:

- Manual user action
- AI recommendation (user approval required)
- Workflow automation (Module 5)

Deal creation is:

- Idempotent
 - Logged
 - Permission-controlled
-

6. Deal Ownership & Assignment

6.1 Ownership Rules

- Each deal has exactly one owner
 - Owner must be a human user
 - AI cannot own deals
-

6.2 Reassignment Triggers

- Manual reassignment
- Stage change
- Inactivity timeout
- Manager override

All reassignments are logged.

7. Stage Progression Rules

7.1 Manual Progression

- Users move deals between stages
 - Permission-gated
 - Reason required for:
 - Skipping stages
 - Moving backward
-

7.2 AI-Assisted Progression

AI may:

- Suggest next stage
- Recommend actions
- Flag stalled deals

AI may **not**:

- Auto-close deals
- Mark deals as Won/Lost without human action

Human-in-the-loop is mandatory.

8. Tasks & Next Actions

- Deals can have associated tasks:
 - Call
 - Follow-up
 - Meeting
- Tasks may be:
 - User-created
 - AI-suggested
 - Workflow-generated

Tasks affect deal health scoring.

9. Win / Loss Management

9.1 Deal Closure

- Closing a deal requires:
 - Status (Won / Lost)
 - Mandatory reason
 - Loss reasons are configurable
-

9.2 Post-Closure Actions

- Trigger workflows
 - Notify stakeholders
 - Archive or recycle lead
-

10. AI Insights on Deals

AI provides:

- Deal health score
- Risk flags
- Suggested next steps
- Objection summaries (from calls)

AI insights are **advisory**, not authoritative.

11. Activity Timeline Integration

Deal timeline aggregates:

- CRM actions
- AI calls
- Tasks
- Notes
- Stage changes

Timeline is immutable and auditable.

12. Permissions & Controls

- Deal access controlled by:
 - Role
 - Ownership
 - Pipeline
 - Managers can:
 - View team deals
 - Override assignments
 - Cross-pipeline access is configurable
-

13. Non-Functional Requirements (Module-Specific)

- Deal updates $\leq 200\text{ms}$ latency
 - Pipeline changes propagate immediately
 - Support ≥ 1 million deals per tenant
 - No data loss on concurrent updates
-

14. Out of Scope (Explicit)

- Revenue forecasting
 - Commission tracking
 - Contract management
-

Module 6 Revised

MODULE 6 — Opportunity, Deals & Revenue Tracking

Version: 1.0

Status: Draft → Pending Confirmation

System: Omni-Channel Martech CRM Platform

Audience: Product, Sales Ops, Finance (read-only), Backend, Analytics, QA

1. Module Overview & Strategic Intent

1.1 Purpose

This module tracks **monetizable intent** after lead qualification and converts CRM activity into **measurable revenue outcomes**.

If:

- Module 3 = Leads (raw demand)
- Module 4 = Conversations
- Module 5 = Orchestration

Then **Module 6** answers one question:

“Did all of that effort actually make money?”

This module is:

- Sales-accurate
 - Pipeline-driven
 - Forecast-aware
 - AI-observable (not AI-controlled)
-

2. Scope

2.1 In Scope

- Opportunity creation & management
- Deal lifecycle tracking
- Revenue estimation & forecasting

- Product & pricing association
- Win/loss analysis
- Attribution to campaigns, AI calls, and channels

2.2 Explicitly Out of Scope

- Invoicing & billing (Module 9)
 - Accounting integrations
 - Commission calculation (future enhancement)
-

3. Opportunity Entity Definition

3.1 Opportunity Identity

Each opportunity must have:

- Opportunity ID (system-generated)
 - Opportunity Name
 - Linked Lead ID (mandatory)
 - Linked Company / Account (optional)
 - Created timestamp
 - Created by (user / system / AI)
-

3.2 Core Attributes

Mandatory:

- Pipeline
- Current Stage
- Owner (user)
- Expected Close Date
- Currency
- Estimated Deal Value

Optional:

- Probability %
- Deal Type (new / renewal / upsell) →
- Source campaign
- Notes
- Tags

4. Opportunity Pipelines & Stages

4.1 Pipelines

- Multiple pipelines per tenant (confirmed)
- Pipelines represent sales motions

Examples:

- New Business
- Renewals
- Enterprise Sales
- Channel Partner Deals

4.2 Stages

Stages are tenant-defined and may include:

- Qualification
- Proposal Sent
- Negotiation
- Verbal Commit
- Won
- Lost

Each stage supports:

- Entry rules
- Exit rules
- Allowed actions
- Required fields (stage gating)

Stage changes must be logged and attributable.

5. Opportunity Creation Rules

5.1 Creation Triggers

Opportunities may be created:

- Manually by users
- Automatically on lead stage transition
- By AI recommendation (approval required)
- Via campaign outcomes

AI **cannot auto-create opportunities without human approval.**

5.2 Lead ↔ Opportunity Relationship

- One lead may have multiple opportunities (configurable)
 - One opportunity belongs to exactly one lead
 - Lead remains active after opportunity creation
-

6. Product, Pricing & Deal Structure

6.1 Deal Line Items

Each opportunity may include:

- Product / service
- Quantity
- Unit price
- Discount (absolute or %)
- Taxes (optional, informational)

Subtotal and total value must auto-calculate.

6.2 Pricing Rules

- Price lists are tenant-configurable
- Discounts beyond threshold may require approval
- All pricing changes logged

This module tracks **deal value**, not invoiced value.

7. Revenue Forecasting

7.1 Forecast Models

Forecasts may be computed using:

- Stage probability weighting
- Manual probability override
- Time-based close projections

Forecast logic must be transparent and configurable.

7.2 Forecast Views

Supported views:

- By owner
- By pipeline
- By stage
- By month / quarter

Forecast data is **read-only** for non-sales roles.

8. AI Assistance (Advisory Only)

8.1 AI Capabilities

AI may:

- Suggest next best action
 - Predict win probability
 - Highlight stalled deals
 - Recommend follow-ups
 - Flag risk signals (silence, objections)
-

8.2 AI Constraints

AI **must not**:

- Change deal value
- Move opportunity stages
- Mark deals as won/lost
- Apply discounts

AI is advisory, not authoritative.

9. Win / Loss Management

9.1 Deal Closure

When an opportunity is closed:

- Status must be Won or Lost
 - Close reason required
 - Timestamp recorded
 - Final deal value locked
-

9.2 Win/Loss Reasons

Tenant-defined reason codes such as:

- Price
- Competition
- Timing
- No budget
- Internal decision

Reasons feed analytics and AI learning loops.

10. Attribution & Traceability

Each opportunity must store attribution links to:

- Lead source
- Campaign(s)
- AI calls
- WhatsApp / SMS / Email interactions

- Human activities

This enables **ROI measurement per channel and per AI action**.

11. Activity Timeline Integration

Opportunity-level timeline includes:

- Stage changes
- Notes
- Proposal sent
- Pricing changes
- Approval events
- Closure

Timeline is immutable and auditable.

12. Search, Filters & Views

Users can filter opportunities by:

- Pipeline
- Stage
- Owner
- Deal value range
- Close date
- Probability
- Tags

Saved views supported per role.

13. Permissions & Security

- Opportunity access governed by RBAC (Module 2)
- Only owners or permitted roles may edit
- No cross-tenant visibility
- All critical actions logged

14. Data Outputs (For Analytics & Billing)

This module emits:

- Revenue projections
- Win/loss data
- Deal velocity metrics
- Channel & campaign attribution
- AI influence signals

Visualization handled in **Module 7**.

Billing linkage handled in **Module 9**.

Closing Position

This module ensures the platform does **not become an activity theater**.

It answers:

- What is selling?
- What is stuck?
- What is real?
- What is noise?

With Module 6 in place, the CRM now connects:

Conversation → Campaign → Opportunity → Revenue

Document design and management

Report design

MODULE 7 — Analytics & Reporting

MODULE 7 — Analytics & Reporting

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module provides **visibility, control, and accountability** across the entire platform. If Modules 3–6 run the business, Module 7 explains **why it's working or failing**.

This is not vanity analytics. This is **decision-grade intelligence** for operators, managers, and executives.

2. Scope

Applies to:

- AI calling performance
- CRM effectiveness
- Campaign efficiency
- Deal progression metrics
- User & agent performance
- Cost transparency

Does **not** include:

- Billing invoices (Module 9)
 - Raw system logs (Module 11)
 - Predictive revenue forecasting (explicitly excluded)
-

3. Analytics Architecture (Logical)

Analytics must support:

- **Real-time dashboards**
- **Daily aggregated snapshots**

- **Historical trend analysis**

Data sources:

- CRM events
- AI call logs
- Campaign execution logs
- Deal lifecycle events
- User activity logs

Analytics must be **read-optimized** and **non-blocking** to core operations.

4. Dashboard Types (Confirmed)

4.1 Platform-Level Dashboards (Super Admin)

- Total call minutes consumed
 - Active tenants
 - AI call success rate
 - Cost vs usage overview
 - Error & failure rates
-

4.2 Tenant-Level Dashboards (Tenant Admin)

- AI vs human call performance
 - Lead conversion funnel
 - Campaign effectiveness
 - Cost per lead
 - Cost per successful call
-

4.3 Role-Based Dashboards

- Manager view:
 - Team performance
 - Deal velocity
 - Stalled opportunities
- Agent view:
 - Assigned leads
 - Tasks

- Call outcomes

Dashboards are permission-aware.

5. Core Metrics (Confirmed)

5.1 AI Calling Metrics

- Call connection rate
 - Call completion rate
 - Average call duration
 - AI containment rate (no human handoff)
 - AI failure rate
-

5.2 CRM & Lead Metrics

- Leads by source
 - Lead-to-call ratio
 - Lead-to-deal conversion
 - Average response time
 - Follow-up compliance
-

5.3 Campaign Metrics

- Campaign reach
 - Success rate per campaign
 - Retry effectiveness
 - Drop-off points
 - A/B test comparison
-

5.4 Deal Metrics

- Deals by stage
- Stage aging
- Win/Loss ratio
- Deal cycle time

5.5 Cost Metrics (Critical)

- Cost per minute
- Cost per lead
- Cost per deal
- Cost per successful outcome

Cost visibility is **mandatory**, not optional.

6. Dashboard Customization (Confirmed: High)

Tenant admins can:

- Add/remove widgets
- Configure date ranges
- Apply filters
- Save dashboard views
- Share dashboards (read-only)

No-code customization only. No scripting.

7. Filters & Segmentation

Supported filters:

- Date range
- Campaign
- AI agent
- Pipeline
- User
- Lead source
- Language

Filters must be:

- Combinable
- Performant
- Persistable

8. Real-Time vs Snapshot Reporting

8.1 Real-Time Dashboards

- Live call counts
- Active campaigns
- Current queue depth

Latency tolerance: ≤ 5 seconds.

8.2 Snapshot Reports

- Daily summaries
- Weekly trends
- Monthly exports

Snapshots are immutable once generated.

9. Exports & External Consumption (Confirmed)

Supported export formats:

- CSV
- JSON (API)
- Scheduled reports

BI-readiness:

- Clean schema
- Stable metric definitions
- Versioned metrics

No direct DB access for customers.

10. Alerts & Thresholds

Users can configure alerts for:

- Usage spikes
- Cost overruns
- Campaign failures
- Drop in conversion rates

Alerts via:

- In-app notifications
- Email

No SMS alerts in v1.

11. Data Accuracy & Trust Rules

- Metrics must be:
 - Deterministic
 - Reproducible
 - Explainable
- Any derived metric must expose:
 - Formula
 - Inputs
 - Time window

No black-box numbers.

12. Permissions & Data Access

- Users only see data they are permitted to see
- Managers see team-level aggregates
- Tenant Admin sees tenant-wide data
- Super Admin sees platform-wide data

No cross-tenant visibility.

13. Non-Functional Requirements (Module-Specific)

- Dashboard load time ≤ 2 seconds
 - Export generation ≤ 30 seconds (large datasets)
 - Must support:
 - ≥ 5 years historical data
 - Analytics system failure must not affect calling or CRM
-

14. Out of Scope (Explicit)

- Predictive analytics
 - ML-driven forecasting
 - Custom SQL access
 - Data warehousing for customers
-

MODULE 8 — Integrations

MODULE 8 — Integrations

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module defines **how the platform connects to the outside world**—cleanly, securely, and without leaking control.

The guiding principle here is **centralized power, controlled exposure**:

- Platform admin owns core infrastructure integrations
- Customers consume capabilities, not credentials

This protects margin, reliability, and compliance.

2. Scope

Applies to:

- Telephony integrations (platform-managed)
- Calendar synchronization
- Webhooks (outbound & inbound)
- Internal vs external integration boundaries

Does **not** include:

- Customer-managed telephony providers
 - Third-party CRM sync
 - Automation marketplaces (Zapier / n8n)
-

3. Telephony Integration (Platform-Managed)

3.1 Ownership Model (Confirmed)

- Telephony providers are:
 - Integrated
 - Configured
 - Maintained**only by Platform Admin**
- Customers **do not** supply provider credentials.
- Customers **select or purchase numbers** from available inventory.

This ensures:

- Cost control
 - Uniform call quality
 - Central compliance enforcement
-

3.2 Phone Number Management

Platform Admin capabilities:

- Procure numbers from providers
- Assign numbers to tenants
- Configure:
 - Inbound routing
 - Outbound caller ID
 - Regional availability
- Set pricing per number (optional)

Tenant capabilities:

- View available numbers
 - Purchase / assign number(s)
 - Release numbers
 - View cost implications per number
-

3.3 Cost Attribution

- Each number is tagged with:
 - Region
 - Cost per minute (inbound / outbound)
- Call cost calculation considers:
 - Number used
 - Direction

- Duration
- Costs feed directly into:
 - Usage metering
 - Analytics
 - Billing (Module 9)

No flat-rate masking. Full transparency.

3.4 Routing Rules

Inbound routing supports:

- AI-first handling
- Time-based routing
- Language-based routing
- Human fallback

Outbound routing supports:

- Campaign-driven number selection
 - Region matching
 - Compliance constraints
-

4. Calendar Integration (Confirmed)

4.1 Supported Calendars

- Google Calendar
 - Microsoft Outlook Calendar
-

4.2 Use Cases

- AI scheduling appointments
 - Availability checking during calls
 - Booking confirmations
 - Rescheduling / cancellation
-

4.3 Sync Behavior

- Two-way sync
- Conflict detection
- Timezone normalization
- Event ownership mapping

Calendar sync must be:

- Explicitly authorized
 - Revocable
 - Logged
-

4.4 AI Boundaries

AI may:

- Check availability
- Create events
- Update events

AI may **not**:

- Delete calendar accounts
 - Access unrelated calendars
-

5. Webhooks

5.1 Outbound Webhooks (Confirmed)

Platform emits webhooks for:

- Lead created / updated
- Call completed
- Campaign completed
- Deal stage changed
- Appointment booked

Webhook payloads are:

- Versioned

- Signed
 - Retry-enabled
-

5.2 Inbound Webhooks (Controlled)

Inbound webhooks allowed for:

- Lead creation
- Lead updates
- Campaign triggers

Constraints:

- Authentication required
- Rate-limited
- Schema-validated

No arbitrary execution allowed.

6. Internal vs External Integration Boundaries

6.1 Internal Integrations

Internal-only:

- Telephony providers
- AI engines
- Billing meters
- Usage trackers

These are **never exposed** to customers.

6.2 External Integrations

Customer-visible:

- Calendar connectors
- Webhooks
- Export APIs (read-only, analytics)

Customers interact with:

- Abstracted endpoints
 - Controlled schemas
 - Limited scopes
-

7. API Exposure Rules

- No raw provider credentials exposed
- No write access to billing or telephony
- All APIs are:
 - Tenant-scoped
 - Permission-aware
 - Rate-limited

API keys:

- Generated by platform
 - Rotatable
 - Revocable
-

8. Error Handling & Resilience

- Provider failures must:
 - Fail gracefully
 - Trigger alerts
 - Pause dependent workflows
- Automatic retries with backoff
- Circuit breakers for unstable providers

No cascading failures allowed.

9. Audit & Observability

- Log:
 - Integration calls
 - Failures
 - Authorization events

- Tenant visibility:
 - Success/failure counts
 - Admin visibility:
 - Provider health
 - Latency
 - Error rates
-

10. Non-Functional Requirements (Module-Specific)

- Webhook delivery SLA $\geq 99.9\%$
 - Calendar sync latency ≤ 10 seconds
 - Telephony routing decision $\leq 200\text{ms}$
 - Integration outages must not corrupt CRM data
-

11. Out of Scope (Explicit)

- Customer-provided telephony credentials
 - Third-party CRM integrations
 - Marketplace connectors
 - Custom customer scripts
-

MODULE 9 — Billing, Usage & License Enforcement

MODULE 9 — Billing, Usage & License Enforcement

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module is the **monetization and financial control layer** of the platform.
Its job is brutally simple and non-negotiable:

- Track every billable unit
- Convert usage into money
- Prevent revenue leakage
- Enforce limits without ambiguity

AI calls cost real money per second.
This module ensures **every second is accounted for**.

2. Scope

Applies to:

- Usage metering
- Per-minute billing
- License validity enforcement
- Invoicing
- Payment gateway integration
- Overuse prevention

Does **not** include:

- Pricing strategy decisions
 - Discounts & coupons (optional future)
 - Revenue forecasting
-

3. Billing Model (Confirmed)

3.1 Pricing Basis

- Primary billable metric: **Credits**
- Billing granularity:
 - Per-second tracking
 - Rounded to per-minute for invoicing (configurable)

No per-seat pricing.

No feature-tier pricing.

3.2 Cost Components

Each call cost is derived from:

- Phone number region
- Direction (inbound / outbound)
- Duration
- AI processing overhead (optional markup)

Costs are **transparent and itemized**.

4. Usage Metering

4.1 Metered Events

- Call started
- Call connected
- Call ended
- AI processing time
- Human handoff (AI billing stops immediately)

Metering data must be:

- Immutable
- Time-stamped
- Attributed to:
 - Tenant
 - Campaign

- AI agent
 - Phone number
-

4.2 Real-Time Usage Tracking

- Live minute consumption counter
- Tenant-visible usage dashboard
- Near-real-time balance updates (≤ 5 seconds lag)

No delayed surprises.

5. License Validity & Enforcement

5.1 License States

- Active
 - Expired
 - Suspended
 - Depleted (usage exhausted)
-

5.2 Enforcement Rules (Confirmed)

- **Hard stop** when usage balance reaches zero:
 - Outbound calls blocked
 - Inbound calls receive system response
 - Campaigns auto-paused
- No grace usage
- No silent overdrafts

This rule is enforced at **API + execution layer**, not UI.

6. Invoicing

6.1 Invoice Generation

Invoices include:

- Billing period
- Total minutes consumed
- Cost breakdown by:
 - Number
 - Campaign
 - Call type
- Taxes (configurable)
- Payment status

Invoices are:

- Auto-generated
 - Immutable once issued
 - Downloadable (PDF)
-

6.2 Invoice Frequency

- Monthly (default)
 - Custom periods supported (admin-configured)
-

7. Payment Gateway Integration (Confirmed: External)

7.1 Supported Gateways

- Stripe
- Razorpay

Gateway selection is:

- Platform-level
 - Tenant-agnostic
-

7.2 Payment Flow

- Invoice generated
- Payment link issued
- Payment confirmation webhook
- License renewed / usage credited

Failed payments trigger:

- Alerts
 - Grace period (optional)
 - Suspension (admin-configured)
-

8. Prepaid vs Postpaid (Clarification Required)

Assumption (Default): Prepaid Usage Model

- Tenant prepays for minutes
- Balance deducted in real time
- Prevents credit risk

Postpaid can be introduced later but is **out of scope for v1 unless you override this assumption**.

9. Alerts & Notifications

Configurable alerts for:

- Low balance thresholds
- High usage spikes
- Invoice generated
- Payment failures

Delivery channels:

- In-app
 - Email
-

10. Admin Controls

Super Admin can:

- Adjust balances (with audit reason)
- Override invoices
- Pause billing

- Issue credits
- View platform-wide revenue

All admin actions are:

- Logged
 - Auditable
 - Non-deletable
-

11. Audit & Compliance

- Complete billing audit trail
- Reconciliation-ready reports
- Tax-compliant invoice numbering
- GDPR-safe storage of billing data

Billing data is retained per legal requirements.

12. Non-Functional Requirements (Module-Specific)

- Metering accuracy $\geq 99.99\%$
- Billing operations must not delay call execution
- Invoice generation ≤ 10 seconds
- Payment confirmation handling ≤ 5 seconds

Failure here is existential—this module must be rock-solid.

13. Out of Scope (Explicit)

- Discounts & promotions
 - Wallet transfers between tenants
 - Manual invoicing by customers
 - Multi-currency pricing (future)
-

MODULE 10 — Admin & Super Admin Controls

MODULE 10 — Admin & Super Admin Controls

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module defines **who ultimately controls the platform** and how risk is contained when things go wrong.

Admin controls are not “settings screens.”

They are **governance mechanisms**—designed to protect revenue, compliance, uptime, and reputation.

This module draws a **hard line** between:

- Tenant autonomy
 - Platform sovereignty
-

2. Scope

Applies to:

- Super Admin (platform owner) controls
- Tenant Admin controls
- Override & emergency actions
- Platform-wide visibility
- Abuse prevention

Does **not** include:

- Day-to-day CRM operations
 - Customer support tooling
 - Analytics dashboards (Module 7)
-

3. Admin Roles (Hierarchy)

3.1 Super Admin (Platform Owner)

Scope:

- Entire platform
- All tenants
- All infrastructure-facing controls

This role is **non-delegable** and **non-customizable**.

3.2 Tenant Admin (Customer)

Scope:

- Single tenant only
- No platform-level access
- No infrastructure access

Tenant Admin is powerful—but **sandboxed**.

4. Super Admin Capabilities (Platform-Level)

4.1 Tenant Governance

- Create / suspend / terminate tenants
- Override tenant status (emergency)
- View tenant health:
 - Usage
 - Errors
 - Cost burn
- Lock tenant access for compliance violations

No tenant action can block Super Admin authority.

4.2 License & Billing Overrides

- Adjust usage balances
- Issue credits / debits
- Pause billing
- Force invoice regeneration
- Override payment failures (with reason)

All overrides require:

- Reason entry
 - Immutable audit log
-

4.3 Telephony & Integration Control

- Manage telephony providers
- Control number inventory
- Enable / disable regions
- Blacklist numbers
- Emergency call shutdown

This is a **kill-switch zone**.

4.4 AI & Campaign Safety Controls

- Global AI throttles
- Max concurrent call limits
- Emergency stop for:
 - Campaigns
 - AI agents
 - Entire platform
- AI behavior override (guardrails only)

Used sparingly. Logged always.

5. Tenant Admin Capabilities (Tenant-Level)

5.1 User & Access Management

- Create / deactivate users
- Assign roles

- Configure permissions (within allowed scope)
- Force logout users

Cannot escalate beyond tenant boundary.

5.2 Operational Controls

- Pause / resume campaigns
- Configure workflows
- Assign numbers
- Manage pipelines
- Configure alerts

Tenant Admin **cannot**:

- Modify billing logic
 - Access telephony credentials
 - See other tenants
-

6. Override & Emergency Actions

6.1 Emergency Scenarios

- Fraudulent call behavior
 - Runaway campaigns
 - Billing anomalies
 - Compliance breach
 - Provider outage
-

6.2 Emergency Toolkit

Super Admin can:

- Stop all calls instantly
- Disable specific tenants
- Lock specific AI agents
- Freeze billing

Actions take effect **immediately**, bypassing normal workflows.

7. Audit & Accountability

7.1 Audit Log Coverage

Logged events include:

- Admin overrides
- License changes
- Tenant suspension
- Emergency actions
- Configuration changes

Audit logs are:

- Immutable
- Time-stamped
- Actor-attributed

7.2 Visibility Rules

- Super Admin: full visibility
- Tenant Admin: tenant-only visibility
- No audit log deletion allowed

8. Configuration Management

8.1 Platform Configurations

- Feature flags
- Regional availability
- Global limits
- Compliance rules

Configs are:

- Versioned
- Rollback-capable
- Change-controlled

8.2 Tenant Configurations

- Business hours
- Data retention
- Notification preferences
- Allowed workflows

Tenant configs cannot override platform constraints.

9. Abuse Prevention & Safeguards

- Call rate anomaly detection
- Cost spike alerts
- Automated campaign pausing
- Repeated failure pattern detection

System must protect **you** even from your own customers.

10. Non-Functional Requirements (Module-Specific)

- Admin actions propagate ≤ 2 seconds
 - Emergency kill switch $\leq 500\text{ms}$
 - Admin UI access requires elevated security
 - Zero tolerance for partial execution of admin actions
-

11. Out of Scope (Explicit)

- Customer support ticketing
 - Delegated super-admin roles
 - Customer-visible platform settings
 - External IAM tools
-

MODULE 11 — Security, Compliance & Audit

MODULE 11 — Security, Compliance & Audit

SRS v1.0 | Status: Draft → Pending Your Final Confirmation

1. Purpose

This module ensures the platform is **defensible, auditable, and enterprise-safe**.

Security is not a feature.

Compliance is not a checkbox.

Auditability is not optional.

This module exists to answer three questions with confidence:

- Can data be trusted?
 - Can misuse be proven?
 - Can regulators, enterprises, and auditors be satisfied?
-

2. Scope

Applies to:

- Data security
- Access control enforcement
- Regulatory compliance
- Audit trails
- Risk containment

Does **not** include:

- Legal policy drafting
 - External certifications (ISO, SOC) execution
 - Customer-side compliance responsibilities
-

3. Security Architecture (Baseline)

3.1 Data Isolation (Confirmed)

- **Dedicated database per tenant**
- No shared schemas
- No cross-tenant queries
- Infrastructure-level separation

This is non-negotiable and already frozen.

3.2 Data Encryption

- Encryption at rest:
 - Databases
 - Call recordings
 - Transcripts
- Encryption in transit:
 - TLS for all APIs
 - Secure webhook delivery

Keys are:

- Rotatable
 - Access-controlled
 - Never exposed to tenants
-

4. Authentication & Authorization

4.1 Authentication

- Email + password authentication
- Password policy enforced platform-wide
- Secure reset flows
- Session expiration & invalidation

No SSO in v1 (confirmed).

4.2 Authorization

- Role-based access control (RBAC)
- Permission checks at:
 - API layer
 - Service layer
- No UI-only enforcement

Authorization failures are:

- Logged
 - Non-destructive
 - Non-retriable
-

5. AI-Specific Security Controls

5.1 AI Guardrails

AI is constrained by:

- Campaign rules
- Workflow permissions
- Hard-coded forbidden actions

AI cannot:

- Access billing systems
 - Modify security settings
 - Access other tenants' data
 - Execute arbitrary code
-

5.2 Prompt & Context Safety

- Prompt templates are versioned
 - Context injection is deterministic
 - No user-controlled raw prompt execution
 - Sensitive fields masked where required
-

6. Call Recording & Voice Data Compliance

6.1 Consent Management (Confirmed)

- Explicit consent capture per call
- Consent stored with:
 - Timestamp
 - Call ID
 - Jurisdiction (if applicable)

Calls without consent:

- Can be blocked
 - Or recorded with redaction (configurable)
-

6.2 Retention Policies

- Tenant-configurable retention windows
- Hard maximums enforced by platform
- Automatic deletion after expiry

Deletion is:

- Logged
 - Irreversible
-

7. Regulatory Compliance (Confirmed)

7.1 GDPR

Supported capabilities:

- Data export (tenant-scoped)
 - Right to deletion
 - Data minimization
 - Purpose limitation
-

7.2 Call & Telecom Compliance

- Jurisdiction-aware call handling
- Recording consent enforcement
- Time-of-day restrictions
- Regional call blocking

Compliance rules override campaign logic.

8. Audit Logging

8.1 What Is Logged (Mandatory)

- Authentication events
 - Authorization failures
 - Role & permission changes
 - Admin overrides
 - Billing adjustments
 - Campaign start/stop
 - AI fallback events
-

8.2 Audit Log Properties

- Immutable
- Time-stamped
- Actor-attributed (user / AI / system)
- Tenant-scoped

Audit logs are **read-only forever**.

9. Monitoring & Incident Detection

9.1 Security Monitoring

- Failed login attempts
 - Abnormal access patterns
 - Permission abuse detection
 - AI misuse signals
-

9.2 Incident Response Hooks

- Automated alerts
- Tenant isolation
- Emergency suspension
- Evidence preservation

No silent incidents allowed.

10. Internal Access Controls

- Least-privilege principle
- Super Admin access logged and reviewed
- No direct database access without justification
- Break-glass access procedures defined

Your own team is also governed.

11. Data Integrity & Reliability

- No hard deletes for operational data
 - Referential integrity enforced
 - Versioning for critical configurations
 - Replayable event logs for dispute resolution
-

12. Non-Functional Requirements (Module-Specific)

- Security checks must not add > 5% latency
 - Audit writes must be async & non-blocking
 - Zero tolerance for cross-tenant leakage
 - Security failures must fail closed, not open
-

13. Out of Scope (Explicit)

- SOC 2 / ISO certification process

- Penetration testing execution
 - Customer-side compliance enforcement
 - Legal document generation
-

Status Tracker

FINAL MODULE LOCK STATUS

Module	Status
Module 1 — Tenant & License Management	✓ Frozen
Module 2 — User & Role Management	✓ Frozen
Module 3 — Lead Management (CRM Core)	✓ Frozen
Module 4 — AI Calling Engine	✓ Frozen
Module 5 — Campaigns & Workflow Automation	✓ Frozen
Module 6 — Deal & Pipeline Management	✓ Frozen
Module 7 — Analytics & Reporting	✓ Frozen
Module 8 — Integrations	✓ Frozen
Module 9 — Billing, Usage & License Enforcement	✓ Frozen
Module 10 — Admin & Super Admin Controls	✓ Frozen
Module 11 — Security, Compliance & Audit	✓ Frozen

WHAT YOU NOW POSSESS

1. Engineering Clarity

- No missing edges
- No “we’ll decide later”
- Clear ownership, boundaries, and failure behavior
- AI autonomy **without** loss of control

Any competent tech lead can now:

- Design architecture
- Break into microservices
- Estimate cost & timeline
- Assign teams without re-asking product questions

2. Business Safety

- Hard billing enforcement
- Platform-owned telephony
- Dedicated tenant isolation
- Kill switches everywhere they should exist

This protects:

- Margins
 - Compliance
 - Reputation
 - Sleep
-

3. Investor / Enterprise Readiness

- Clear monetization logic
- Strong governance
- Auditability baked in
- No cowboy AI behavior

This is the difference between:

“Cool demo”

and

“We can deploy this at scale”



RECOMMENDED NEXT STEPS (Execution Order)

I recommend doing these **in sequence**, not parallel:

Step 1 — Architecture Blueprint

- Logical architecture
- Service boundaries
- Data ownership per service
- Telephony & AI orchestration flow

Step 2 — Dev Phasing Plan

- Phase 1: Revenue-critical path (Modules 1, 3, 4, 5, 9)
- Phase 2: Scale & insight (Modules 6, 7)
- Phase 3: Hardening & governance (Modules 8, 10, 11)

Step 3 — Cost & Infra Model

- AI compute sizing
- Telephony cost modeling
- Per-minute margin analysis
- Break-even math

Step 4 — Engineering Handoff Pack

- SRS (this)
- Architecture diagrams
- API principles
- Non-negotiable constraints



HARD TRUTH (Worth Saying)

Most AI calling products fail because:

- Billing is soft
- AI autonomy is sloppy
- Telephony is leaky
- Admin controls are an afterthought

We avoided all four.

This spec is **not optimistic**—it's **defensive and scalable**, which is exactly what survives real customers.

MODULE 12 — User & Role Management

MODULE 12 — User & Role Management

SRS v1.0 | Status: Draft → Pending Your Confirmation

1. Purpose

This module governs **who can do what, where, and how** across the platform. It enforces security, operational control, and accountability without hardcoding roles—critical for enterprise CRM + AI systems.

This is **not a basic user module**. It is a **permission orchestration layer**.

2. Scope

Applies to:

- CRM
 - AI Calling
 - Campaigns
 - Analytics
 - Billing visibility
 - Admin actions
 - White-label identity management - include
-

3. Role Model (Frozen Decisions)

3.1 Role Philosophy

- Roles are **tenant-defined**
- Permissions are **module + action based** (view, new, edit, delete, print, export, email, whatsapp, attachment,)
- No assumption that users = sales agents
- AI agents are **not tied to users**
- **Named user - admin can create 15 users**
- **Concurrent user - only 5 users can use or login at a time**

This avoids the classic CRM mistake of overbinding AI to humans.

4. Default System Roles (Seed Roles)

These are **templates**, not enforced roles.

Role	Description
Super Admin	Platform owner (internal only)
Tenant Admin	Full control within tenant
Manager	Team & performance oversight
Sales Agent	CRM + calling execution
Support Agent	Inbound / support flows
Auditor	Read-only across modules

Tenants can:

- Modify permissions
 - Clone roles
 - Create new roles
 - Delete unused roles (except Super Admin)
-

5. Permission Model (Core Design)

5.1 Permission Structure

Permissions are defined as:

Module → Submodule → Action

Example:

- CRM → Leads → Edit
- AI → Campaigns → Start
- Analytics → Dashboard → View

- Billing → Usage → View

This enables **fine-grained access without code changes**.

5.2 Permission Categories

- View
- Create
- Edit
- Delete
- Execute (calls, campaigns)
- Approve (managerial actions)
- Export
- Configure
- communication/ notification

Permissions are **explicit**, not implied.

6. Custom Role Builder (Confirmed)

6.1 Features

- UI-based role creation
- Toggle permissions per module
- Clone existing role
- Assign multiple roles to a user (optional, configurable)

6.2 Constraints

- No role can override Super Admin controls
 - Billing write-access restricted to Tenant Admin only (hard rule)
-

7. User Management

7.1 User Lifecycle

- Create user
- Assign role(s)

- Activate / deactivate
- Soft delete (audit-safe)

7.2 User Attributes

- Name
 - Email (unique per tenant) → for login
 - 2FA → optional
 - Phone (optional)
 - Role(s)
 - Status
 - Last login
 - Activity summary
-

8. Authentication & Sessions

8.1 Authentication

- Email + password
- Password policy:
 - Min length
 - Complexity rules
- Password reset via email
- Device Based → authorisation → version 2
- 2FA → optional

SSO is **explicitly excluded** in v1.

8.2 Session Management

- Token-based authentication
 - Configurable session timeout →
 - Force logout by Tenant Admin
 - Concurrent session control - included
-

9. Activity Logging & Audit

9.1 Logged Events (detailed document required - Sujal)

- Login / logout
 - Role changes
 - Permission changes
 - Campaign start/stop
 - Manual CRM edits
 - AI overrides by humans
1. Audit table against each module to save all actions with filters
 2. Another options to do it

9.2 Visibility

- Tenant Admin can view tenant-level logs
 - Super Admin can view platform-wide logs
 - Logs are **read-only**
-

10. AI Agent vs User Boundary (Critical)

10.1 Confirmed Behavior

- AI agents:
 - Are **system entities**
 - Act based on campaign + rules
 - Do NOT belong to users

10.2 Attribution

- Every AI action is tagged with:
 - AI Agent ID
 - Campaign ID
 - Tenant ID

This prevents blame ambiguity and simplifies analytics.

11. Security Rules (Hard Constraints)

- Users cannot escalate their own roles
- Users cannot modify roles they are assigned to
- Tenant Admin cannot access Super Admin features
- No cross-tenant user visibility

Violations must be blocked at **API level**, not UI only.

12. Non-Functional Requirements (Module-Specific)

- Role permission checks must add $\leq 50\text{ms}$ overhead
 - Permission evaluation must be cached
 - Role changes take effect immediately
 - System must support:
 - $\geq 1,000$ users per tenant
-

13. Out of Scope (Explicit)

- SSO / Identity federation
 - Temporary access tokens
 - External IAM integrations
-

Module 2 - V1.1

MODULE 2 — User & Role Management

Version: 1.0

Status: Draft → Pending Confirmation

System: Omni-Channel CRM Platform

Audience: Product, Backend, Security, QA, Compliance

1. Module Overview & Objective

1.1 Purpose

This module governs **who can do what, where, and under which constraints** across the entire platform.

It is designed to:

- Enforce **strict access control**
- Enable **enterprise-grade flexibility**
- Maintain **full accountability and auditability**
- Avoid hardcoded roles and brittle assumptions

This is **not** a basic user system.

It is a **permission orchestration layer** built for AI-driven, multi-module SaaS.

2. Module Scope

This module applies uniformly across:

- CRM (Leads, Contacts, Companies, Deals)
- AI Calling
- Campaign Management
- Analytics & Dashboards
- Billing & Usage Visibility
- Administrative & Configuration Actions
- White-label identity management (included)

Scope enforcement is mandatory at **API level**, not UI level.

3. Role Model (Frozen Decisions)

3.1 Role Philosophy

- Roles are **tenant-defined**
- Permissions are **explicit and granular**
- No system assumption that:
 - Users = sales agents
 - AI agents belong to users
- AI entities and human users are **intentionally decoupled**

This design prevents the classic CRM failure of binding automation to individuals.

3.2 User Type Constraints

- **Named Users**
 - Maximum: 15 per tenant
 - Created and managed by Tenant Admin
 - **Concurrent Users**
 - Maximum: 5 active sessions at a time
 - Enforced at session/authentication layer
 - Excess concurrent logins must be blocked, not queued
-

4. Default System Roles (Seed Roles)

These roles are **templates**, not enforced contracts.

Role	Description
Super Admin	Platform owner, internal only
Tenant Admin	Full control within tenant boundary
Manager	Team oversight, approvals, analytics
Sales Agent	CRM execution + AI calling
Support Agent	Inbound & support workflows
Auditor	Read-only access across modules

Rules

- Tenants may:
 - Modify permissions
 - Clone roles
 - Create new roles
 - Delete unused roles
 - **Super Admin role cannot be modified or deleted**
 - Tenant Admin cannot grant Super Admin privileges
-

5. Permission Model (Core Design)

5.1 Permission Structure

Permissions are defined as:

Module → Submodule → Action

Examples:

- CRM → Leads → Edit
- AI Calling → Campaigns → Start
- Analytics → Dashboard → View
- Billing → Usage → View

This allows:

- Fine-grained access control
 - Permission changes without code deployment
 - Predictable security behavior
-

5.2 Permission Actions (Standardized Set)

Permissions must be explicitly assigned. No implied access.

Supported actions include:

- View
- Create
- Edit
- Delete

- Execute (calls, campaigns, workflows)
 - Approve (managerial actions)
 - Export
 - Configure
 - Communication / Notification
 - Attachment handling
 - Print
-

6. Custom Role Builder (Confirmed)

6.1 Functional Capabilities

Tenant Admin shall be able to:

- Create roles via UI
- Toggle permissions per module/submodule
- Clone existing roles
- Assign multiple roles to a single user (configurable)

Permission evaluation must resolve conflicts deterministically (deny overrides allow).

6.2 Hard Constraints

- No role may override Super Admin authority
 - Billing **write access** is restricted to **Tenant Admin only**
 - This restriction is enforced at API layer
-

7. User Management

7.1 User Lifecycle

Tenant Admin shall be able to:

- Create users
- Assign role(s)
- Activate / deactivate users
- Soft-delete users (audit-safe)

Hard deletion is not allowed via UI.

7.2 User Attributes

Each user record must include:

- Full Name
 - Email (unique per tenant; primary login identifier)
 - Phone number (optional)
 - Assigned role(s)
 - Account status
 - Last login timestamp
 - Activity summary (derived)
-

8. Authentication & Session Management

8.1 Authentication Mechanism

- Email + password login
 - Password policy:
 - Minimum length
 - Complexity rules
 - Password reset via email
 - Device-based authorization (Version 2)
 - Two-factor authentication (optional, configurable per tenant)
 - SSO is explicitly excluded in v1
-

8.2 Session Management

- Token-based authentication
 - Configurable session timeout
 - Tenant Admin can force logout users
 - Concurrent session limit enforced strictly
 - Session violations must terminate oldest session or deny new login (configurable)
-

9. Activity Logging & Audit

9.1 Logged Events

All actions below must be logged immutably:

- Login / logout
- Role assignment or change
- Permission modification
- Campaign start / stop
- Manual CRM edits
- AI overrides initiated by humans

Each module must maintain an **audit table** with:

- Actor (user / system)
- Action
- Timestamp
- Entity affected
- Source (UI / API)

Alternative audit architectures may be proposed separately.

9.2 Log Visibility

- Tenant Admin: tenant-level logs only
 - Super Admin: platform-wide logs
 - Logs are read-only and immutable
-

10. AI Agent vs User Boundary (Critical)

10.1 Confirmed Behavior

- AI agents are system entities
 - AI actions are driven by campaigns and rules
 - AI agents do **not** belong to users
-

10.2 Attribution Rules

Every AI action must be tagged with:

- AI Agent ID
- Campaign ID
- Tenant ID

This ensures:

- Clear accountability
 - Clean analytics
 - No human blame ambiguity
-

11. Security Rules (Hard Constraints)

The system must enforce the following without exception:

- Users cannot escalate their own roles
- Users cannot modify roles assigned to themselves
- Tenant Admin cannot access Super Admin capabilities
- No cross-tenant user visibility
- All violations must be blocked at **API level**

UI-only enforcement is considered a security failure.

Stakeholders

1. Hitech solution Admin → full access
2. Hitech Power Users →
3. Customer Admin → full access
4. Customer Power Users
5. Sales/marketing Team (More mobile friendly)
6. Distributor team (customer ka customer) → customer portal

Story

1. Hitech solution → Company admin
2. Techavtar → customer admin
3. Jio → Distributor

High Leval Summary

SRS — v0.1 (Discovery-in-Progress)

1. Product Overview

1.1 Purpose

The system is a **highly configurable AI Calling SaaS + CRM platform** enabling businesses to:

- Capture leads from multiple channels
- Automatically engage leads via AI voice agents
- Manage conversations, follow-ups, deals, and closures
- Track full customer lifecycle from lead → revenue

1.2 Target Customers

- SMEs & Enterprises
- Sales-driven businesses
- Support & appointment-based businesses
- Agencies reselling AI calling under license

1.3 Business Model

- License-based SaaS (not pure per-seat SaaS)
- Likely tiers based on:
 - Concurrent AI calls
 - Minutes consumed
 - CRM users
 - Feature access
 - White-labeling

 **Decision pending:** resale / white-label vs direct-only.

2. Core System Modules (High-Level)

1. Tenant & License Management
2. User & Role Management
3. Lead Management (CRM Core)
4. AI Calling Engine
5. Campaign & Workflow Automation

6. Conversation & Call Analytics
7. Deal / Pipeline Management
8. Integrations (Telephony, CRM, Calendars, APIs)
9. Billing, Usage & License Enforcement
10. Admin & Super Admin Controls
11. Security, Compliance & Audit

We will lock these one by one.

MODULE 1 — Tenant & License Management

3.1 Description

Supports **multi-tenant architecture** where each company operates in isolated environment with configurable limits enforced by license.

3.2 Functional Requirements (Draft)

- Tenant creation with:
 - Company profile
 - Domain / subdomain
 - Timezone & locale
- License assignment per tenant
- Feature gating based on license
- Usage tracking per tenant:
 - Credits
 - Number of AI agents
 - CRM records
- License expiry, suspension, renewal

3.3 Open Questions (Answer These)

Please respond **inline** or numbered:

1. **License Type**
 - a) Annual / Monthly license?
 - Can be set through admin as monthly, yearly or custom
 - b) Per-company flat license OR usage-based slabs?
 - Usage based slab

2. **White Label**

- a) Do customers get custom branding (logo, colors, domain)?
- no
- b) Is this a paid license tier?
- No - there will be no tier but yes customer will pay per minute call

3. **Environment Isolation**

- a) Logical isolation (shared DB with tenant_id)?
- no
- b) Dedicated DB for enterprise customers?
- Yes

4. **License Enforcement**

- Hard block when exceeded OR soft limit with alerts?
 - Yes
-

MODULE 2 — User & Role Management

4.1 Description

Role-based access control across CRM, AI, analytics, and admin features.

4.2 Draft Roles (Editable)

- Super Admin (Your internal team)
- Tenant Admin
- Manager
- Sales Agent
- Support Agent
- Read-only / Auditor

4.3 Functional Requirements

- Role-based permissions matrix
- Custom roles per tenant
- User activity logging
- Session management
- MFA (optional)

4.4 Questions

5. Do you want **custom role builder** or fixed roles?
Custom role builder

6. Should AI agents be mapped to **specific human users**?
No - users can be assigned module and permission on that module
 7. Any SSO requirement (Google, Azure AD, Okta)?
no
-

MODULE 3 — Lead Management (CRM Core)

5.1 Description

Centralized CRM handling lead intake, enrichment, assignment, and lifecycle.

5.2 Lead Sources (Draft)

- Web forms
- CSV import
- API
- WhatsApp / SMS
- Manual entry
- AI-call generated leads

5.3 Functional Requirements

- Lead stages (configurable)
- Auto-assignment rules
- Lead scoring
- Duplicate detection
- Activity timeline per lead

5.4 Questions

8. Do you want **fully customizable pipelines** per tenant?
Yes
9. Lead scoring:
 - Rule-based
 - AI-based
 - Both?
both
10. Should AI be allowed to **update lead fields automatically**?
Yes with informations and can do follow up

MODULE 4 — AI Calling Engine (CORE DIFFERENTIATOR)

6.1 Description

AI voice agents capable of outbound & inbound calls, integrated tightly with CRM.

6.2 Capabilities (Assumed)

- Natural conversation (interruptions, pauses)
- Context awareness from CRM
- Action execution (booking, updating CRM)
- Call transfer to human
- Multi-language support

6.3 Configuration Levels

- Global AI templates
- Tenant-level AI agents
- Campaign-specific behavior

6.4 Questions (Critical)

11. Outbound, Inbound, or both?
both

12. AI agent types:

- Sales
- Support
- Collections
- Survey

all

13. Should customers **design conversation flows** or only prompt-based?
both

14. Fallback rules when AI fails:

- Retry
- Transfer
- Drop & tag

All and followup as well

MODULE 5 — Campaigns & Automation

7.1 Description

Define when, how, and whom AI should call.

7.2 Features (Draft)

- Campaign scheduler
- Retry logic
- Time-window rules
- Dynamic scripts based on lead data
- A/B testing of AI prompts

7.3 Questions

15. Do campaigns run **event-driven** (new lead) or **batch-based**?
both
 16. Should non-technical users build workflows visually?
Yes with drag and drop features
 17. Is WhatsApp/SMS follow-up part of v1?
yes
-

MODULE 6 — Deal & Pipeline Management

8.1 Description

Manage opportunities post-lead qualification.

8.2 Features

- Deal stages
- Value tracking
- AI-assisted next-step suggestions
- Win/Loss analysis

8.3 Questions

18. Single pipeline or multiple pipelines per tenant?
multiple

19. Revenue forecasting required in v1?
no
-

MODULE 7 — Analytics & Reporting

9.1 Metrics

- Call success rate
- AI vs human conversion
- Cost per lead
- Deal velocity
- Agent performance

9.2 Questions

20. Real-time dashboards or daily snapshots?
both
21. Export requirements (CSV, API, BI tools)?
All and highly configurable dashboard
-

MODULE 8 — Integrations

10.1 Likely Integrations

- Telephony providers
- Calendar (Google, Outlook)
- CRM (optional)
- Webhooks

10.2 Questions

22. Preferred telephony vendors?
No platform owner will provide mobile numbers and customer can buy by themselves
23. Open API for customers mandatory?
No → platform owner will provide all keys and configure. Customer will use and pay per minute

24. Zapier / n8n support?
NO

MODULE 9 — Billing, Usage & Compliance

11.1 Features

- Usage metering
- Invoices
- Alerts
- Compliance logs

11.2 Questions

25. Built-in billing or external (Stripe, Razorpay)?
external

26. Compliance targets:

- GDPR
 - Call recording consent
 - Data retention policies?
- all
-